



LECTURE : 56

12 RESPONSIBLE
CONSUMPTION
AND PRODUCTION



△ 113%



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- **Natural Resources :**

- Natural resources are those that exist in the absence of human intervention. A natural resource can exist as a distinct entity like fresh water, air, or any living thing like a fish, or it can exist in an alternative form that must be processed to acquire the resource like metal ores, rare-earth elements, petroleum, and most types of energy.

- **SDG 12 Advocates for:**

- To reduce ecological footprint by changing the way we produce and consume goods and resources.
- The efficient management of our shared natural resources, and the way we dispose of toxic waste and pollutants;
- Encouraging industries, businesses and consumers to recycle and reduce waste is equally important;
- Creating more efficient production and supply chains to ensure food security and shift us towards a more resource efficient economy.



Facts of today's world

1.3 billion- 1.3 billion tonnes of food is wasted every year, while almost 2 billion people go hungry or undernourished.

22%- The food sector accounts for around 22 percent of total greenhouse gas emissions, largely from the conversion of forests into farmland.

2 billion -Globally, 2 billion people are overweight or obese.

3% - Only 3 percent of the world's water is fresh (drinkable), and humans are using it faster than nature can replenish it.

US\$120 billion -If people everywhere switched to energy efficient lightbulbs, the world would save US\$120 billion annually.

20% - One-fifth of the world's final energy consumption in 2013 was from renewable sources.



CONSUMPTION AND USE OF NATURAL RESOURCES

1. 66% of Electricity is produced daily is from Coal.
2. 23 million Metric Ton of fish is consumed daily all over the world.
3. 97,103,871 barrels of fuel is consumed per day.
4. 11 Million Pounds Of Food consumed every Day.
5. \$33 trillion annually approximately is quarter of the world's population depends on the forest resources.

HOW THEY ARE GETTING DEPLETED ?



CASE STUDY OF WATER POLLUTION IN INDIA

River	Chromium	Lead	Iron
Ganga	Exceeds		Exceeds
Yamuna			Exceeds
Brahmaputra			Exceeds
Ramganga		Exceeds	
Rapti	Exceeds		Exceeds
Narmada			Exceeds
Godavari			Exceeds

The Central Water Commission (CWC) has reported that the samples from two-thirds of the water quality stations spanning India's major rivers which are mentioned below are contaminated by one or more heavy metals, exceeding safe limits set by the Bureau of Indian Standards. Reports says 65% were polluted by heavy metals.

Challenges in
Controlling the
water pollution and
over consumption

CHALLENGES

- Inadequate environmental impact assessments.
- Unsustained water quality monitoring
- Lack of decentralized data generation and participatory data gathering involving all stakeholders.
- Low collaboration and buy-in from private sector.
- Poor communication with industry and other stakeholders
- Inadequate capacity to carry policy commitments, especially monitoring and enforcement.
- Low public awareness on the state of water quality and pollution
- Insufficient access to green technologies
- Lack of funding for monitoring technologies



INDUSTRIAL WASTE: OVERVIEW

Generation of waste is an inevitable consequence of industrial progress and therefore efficient waste management is a matter of concern. There are two main types of industrial waste:

- Hazardous Industrial Waste
- Non-Hazardous Industrial Waste

- no specific disposal sites
- small scale nature and even do not seek consents of SPCBs/PCCs
- Industries are located in non-conforming areas
- No regular interaction between urban local bodies and SPCBs/PCCs to deal such issues

The Central Pollution Control Board (CPCB) an autonomous agency serves as an advisory role to the Government and State Pollution Control Boards (SPCB) in matters relating to the implementation and enforcement of the Air, Water and Environmental Acts. States are monitored by the concerned State Pollution Control Board (SPCB) or Pollution Control Committee (PCC) in Union Territory.

SOLUTIONS FOR NATURAL RESOURCE DEPLETION

1. Make Electricity use more efficient.
2. Use more renewable energy.
3. Promote sustainable fishing growth.
4. Avoid single use plastics.
5. Drive less.
6. Use sustainable agriculture practices.
7. Reduce food waste.
8. Promote sustainable forest management.
9. Treat wastewater before discharging.



GOAL 12. Ensure sustainable consumption and production patterns

TARGET 12.1 Implement the 10-Year Framework of Programmes on Sustainable Consumption and Production Patterns, all countries taking action, with developed countries taking the lead, taking into account the development and capabilities of developing countries

TARGET 12.2 By 2030, achieve the sustainable management and efficient use of natural resources

Global **INDICATOR 12.2.2** Domestic material consumption, domestic material consumption per capita, and domestic material consumption per GDP

Global Metadata

TARGET 12.3 By 2030, halve per capita global food waste at the retail and consumer levels and reduce food losses along production and supply chains, including post-harvest losses

TARGET 12.4 By 2020, achieve the environmentally sound management of chemicals and all wastes throughout their life cycle, in accordance with agreed international frameworks, and significantly reduce their release to air, water and soil in order to minimize their adverse impacts on human health and the environment



Global **INDICATOR 12.4.1** Number of parties to international multilateral environmental agreements on hazardous waste, and other chemicals that meet their commitments and obligations in transmitting information as required by each relevant agreement

Global
Metadata

TARGET 12.5 By 2030, substantially reduce waste generation through prevention, reduction, recycling and reuse

Regional **INDICATOR P-12.5.1** Proportion of recycled waste in relation to total collected waste

TARGET 12.6 Encourage companies, especially large and transnational companies, to adopt sustainable practices and to integrate sustainability information into their reporting cycle

TARGET 12.7 Promote public procurement practices that are sustainable, in accordance with national policies and priorities

TARGET 12.8 By 2030, ensure that people everywhere have the relevant information and awareness for sustainable development and lifestyles in harmony with nature



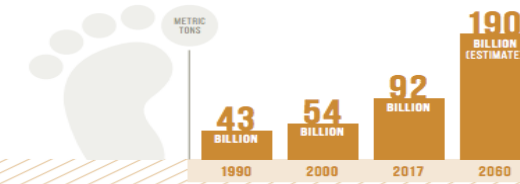
- One of the greatest global challenges is **to integrate environmental sustainability with economic growth and welfare** by **decoupling environmental degradation** from economic growth and doing more with less.
- Resource decoupling and impact decoupling are needed promote sustainable consumption and production (SCP) patterns and to make the transition towards a greener and more socially inclusive global economy.**
- To ensure sustainable consumption and production practices necessarily entails to **respect the biophysical boundaries of the planet** and to **reduce current global consumption rates** in order **to fit with the biophysical capacity to produce ecosystem services and benefits.**

12 RESPONSIBLE CONSUMPTION AND PRODUCTION



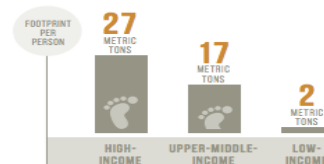
ENSURE SUSTAINABLE CONSUMPTION AND PRODUCTION PATTERNS

THE GLOBAL MATERIAL FOOTPRINT IS RAPIDLY GROWING, OUTPACING POPULATION AND ECONOMIC GROWTH



MATERIAL FOOTPRINT PER CAPITA IN HIGH-INCOME COUNTRIES IS

60% HIGHER THAN IN UPPER-MIDDLE-INCOME COUNTRIES **AND MORE THAN 13 TIMES** THE LEVEL OF LOW-INCOME COUNTRIES



DEVELOPED COUNTRIES **USE ONE FIFTH** OF NATURAL RESOURCES

TO PRODUCE THE SAME AMOUNT OF ECONOMIC OUTPUT AS DEVELOPING COUNTRIES

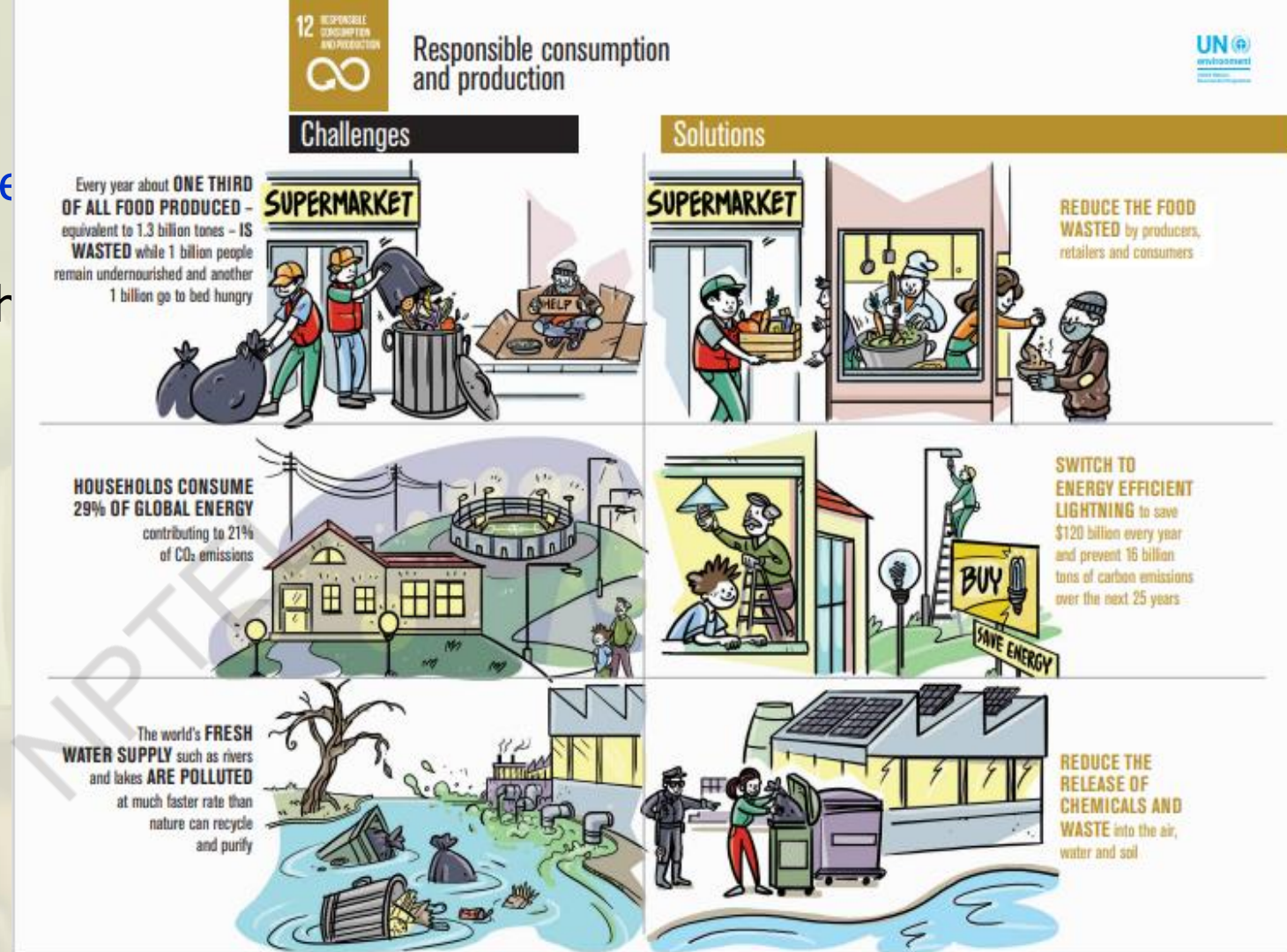
NEARLY 100 COUNTRIES ARE **ACTIVELY** ADOPTING POLICIES AND MEASURES

TO PROMOTE SUSTAINABLE CONSUMPTION AND PRODUCTION

303 POLICIES AND INSTRUMENTS ARE IN PLACE GLOBALLY



- Sustainable growth and development requires **minimizing the natural resource and toxic materials used, and the waste and pollutants generated**, throughout the entire production and consumption process.
- Sustainable Development Goal 12 encourages more **sustainable consumption and production patterns** through various measures, including **specific policies and international agreements** on the management of materials that are toxic to the environment.



- Societies need to find just and equitable ways to meet individual needs and aspirations within the ecological limits of the planet.
- Sustainable practices in the production phase of products and services will not provide sufficient responses to meet science-based emissions reduction targets, natural resource constraints or the growing demand for basic needs such as food, water and sanitation, and access to energy.
- Pollution is the largest environmental cause of disease and premature death globally, responsible for an estimated 9 million premature deaths in 2015 – 16 per cent of all deaths worldwide and three times more deaths than AIDS, tuberculosis and malaria combined.
- ➔ One third of food produced every year is wasted, yet food security is a growing concern.





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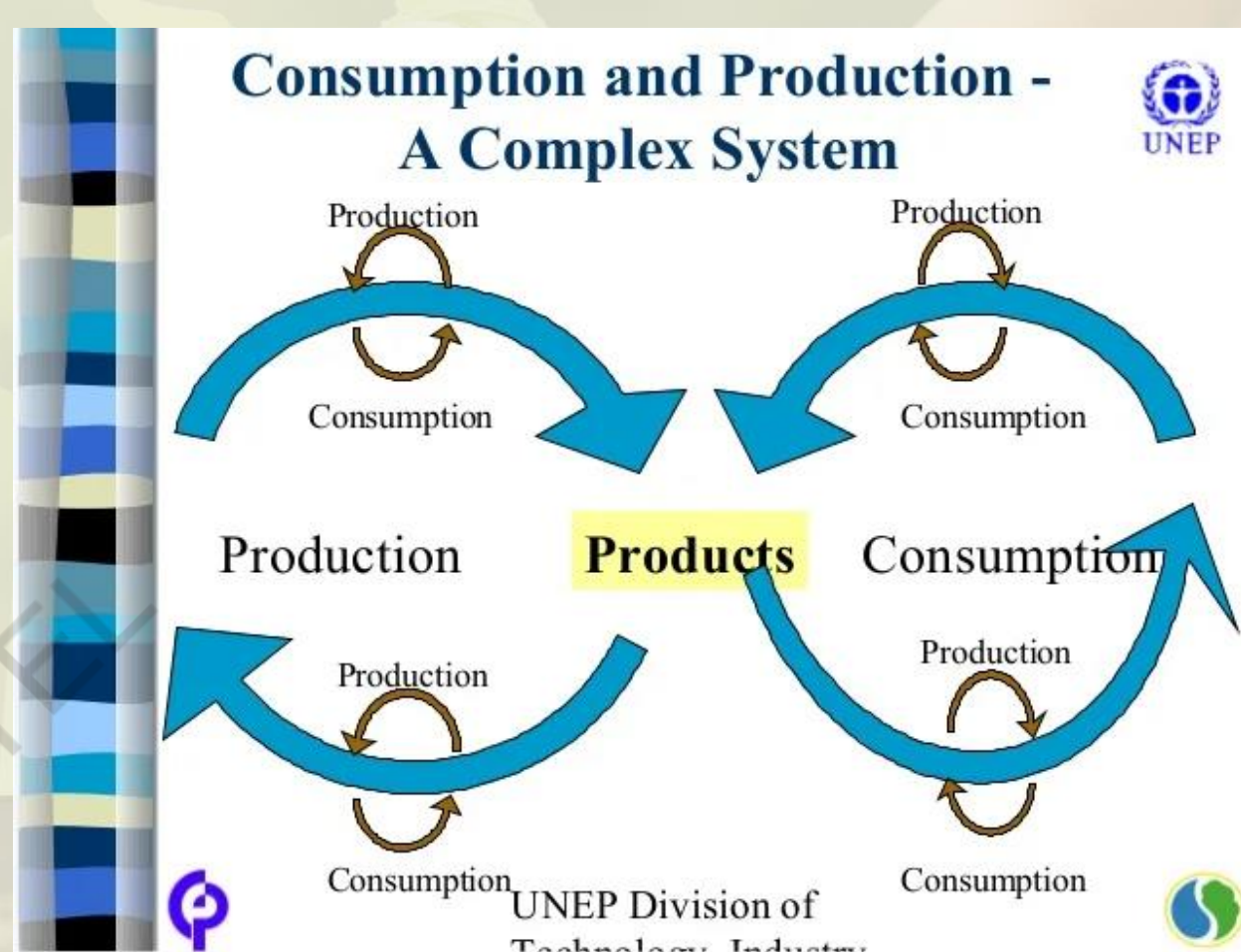
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- **Demand for materials** will likely outpace efficiency gains in supply chains as well as overwhelm **already stretched ecosystem services**.
- **Consumption patterns** need to be made sustainable, particularly **lifestyles** in industrialized societies, and **reduce their ecological footprint** to allow for the **regeneration of natural resources** on which human life and biodiversity depend.
- A **better understanding** of environmental and social impacts of products and services is needed, both **of product life cycles** and **how these are affected by use within lifestyles**.



- ❑ One of the core objectives of the 2030 Agenda for Sustainable Development is **to decouple economic growth from resource use and environmental degradation**, notably through **improved resource efficiency**
- The extraction of primary materials has tripled in the past four decades, which has led to **severe environmental damage and depletion of natural resources**, and **inequalities** have kept growing, both within and between countries.
- SDG 12 promotes SCP, means using services and producing products that **minimize the use of natural resources and toxic materials as well as the emissions of waste and pollutants** over the life cycle of the service or product so as **not to jeopardize the needs of future generations**.



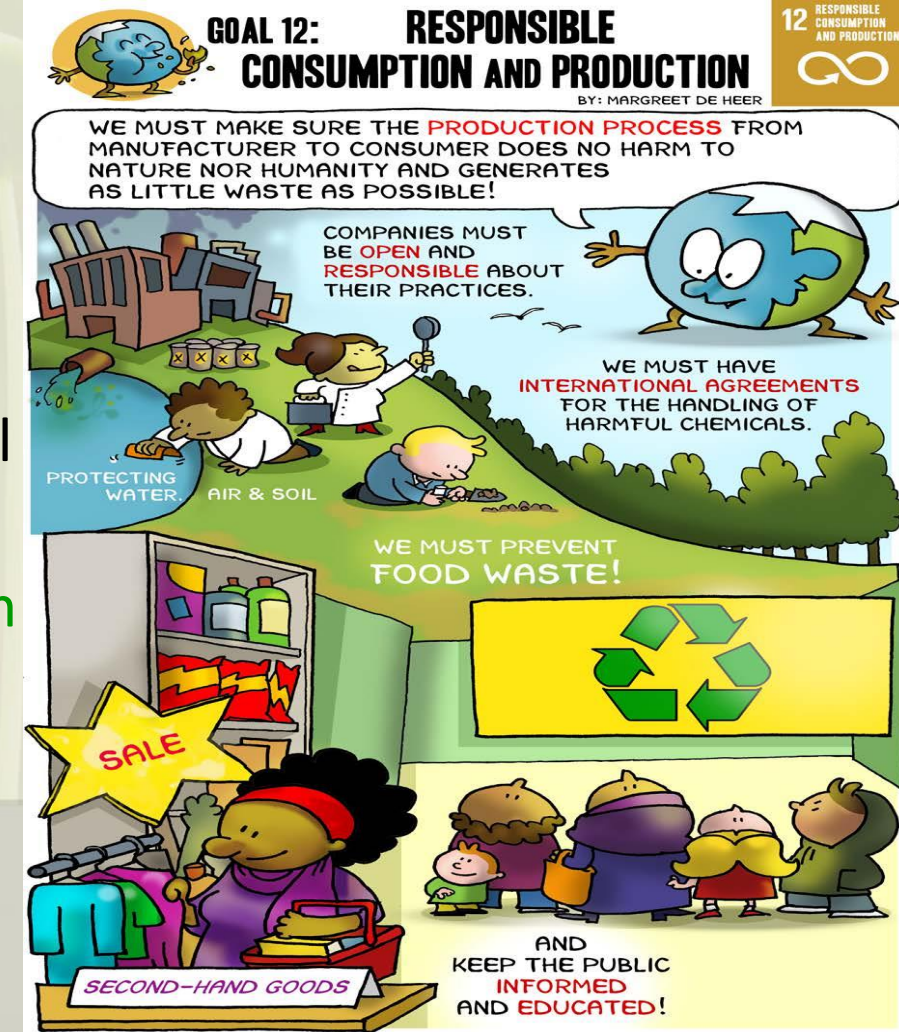
- Identifying “hot spots” within the value chain where interventions have the greatest potential to improve the environmental and social impact of the system as a whole is a crucial first step.

UN Environment and SDG 12

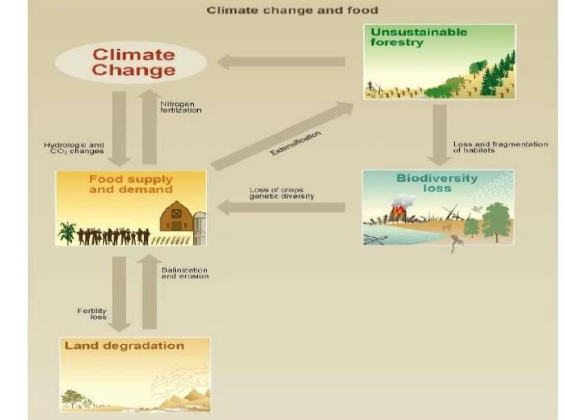
- The UN Environment Programme (UN Environment) has been promoting the shift to SCP patterns for the past two decades; UN work focuses on the areas such as :
 - → Engaging with Governments in their transition to SCP patterns and inclusive green economies
 - → Fostering partnerships with business and industry to promote green investments and apply sustainable business practices across value chains
 - → Disseminating consumer information and promoting awareness and engagement towards sustainable lifestyles



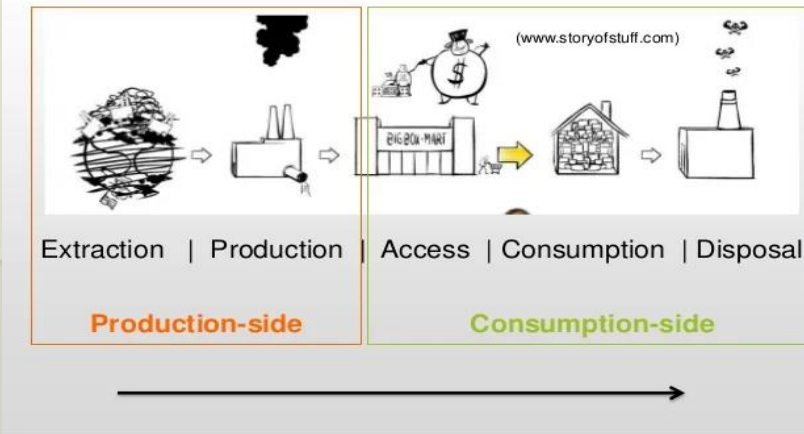
- → Strengthening and communicating the knowledge and scientific base for resource efficiency and SCP
- → Supporting fiscal policy reform to be better aligned with inclusive green economies
- → Strengthening the sound management of chemicals and waste by supporting parties to the multilateral environmental agreements to meet their respective obligations
- → It aims to protect human health and the environment from harmful effects of chemicals and waste by applying the life-cycle approach to the production, use, trade and disposal of chemicals and products containing hazardous substances
- → It promotes the environmentally sound management of hazardous and other wastes and their disposal, including efforts to prevent and minimize their generation and efficiently and safely manage that which cannot be avoided.



- SCP therefore contributes to resilient societies in several ways:
- → Its focus on resource efficiency aims to reduce resource use per unit of GDP, which in turn reduces vulnerability to resource scarcity and resource price volatility, while also ensuring future security of supply
- → It also aims to mitigate environmental degradation that leads to vulnerable societies, such as land degradation, water scarcity, pollution and toxic chemicals, and climatic impacts
- UN Environment collaborates with the private sector in partnerships promoting innovation, technological solutions and financing to tackle our most pressing global environmental challenges;
- As part of its private sector engagement work, it also promotes and builds capacity in corporate sustainability reporting.



PRODUCTION-CONSUMPTION CHAIN



- It is in the interest of business to find **new solutions that enable sustainable consumption and production** patterns;
- A better **understanding of environmental and social impacts of products and services** is needed, both of product life cycles and how these are affected by use within lifestyles;
- **Consumption patterns need to be made sustainable**, particularly lifestyles in industrialized societies, and **reduce their ecological footprint to allow for the regeneration of natural resources** on which human life and biodiversity depend;
- Business can use its **innovative power to then design appropriate solutions** that can both enable and **inspire individuals to lead more sustainable lifestyles**, reducing impacts and improving well-being.



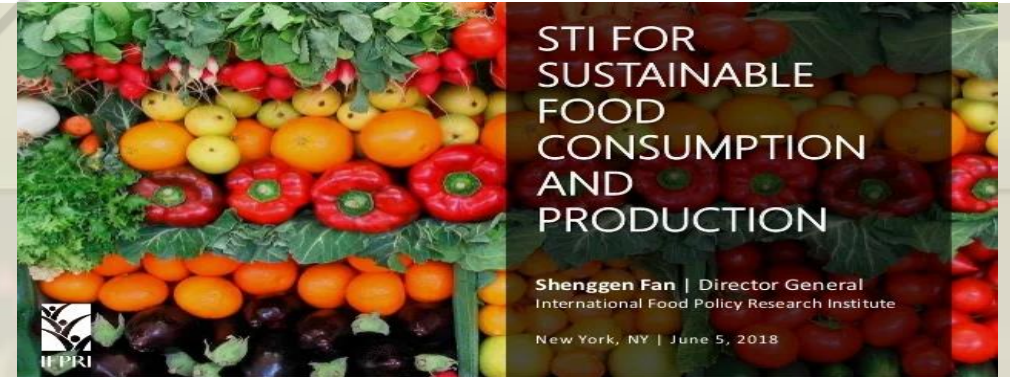
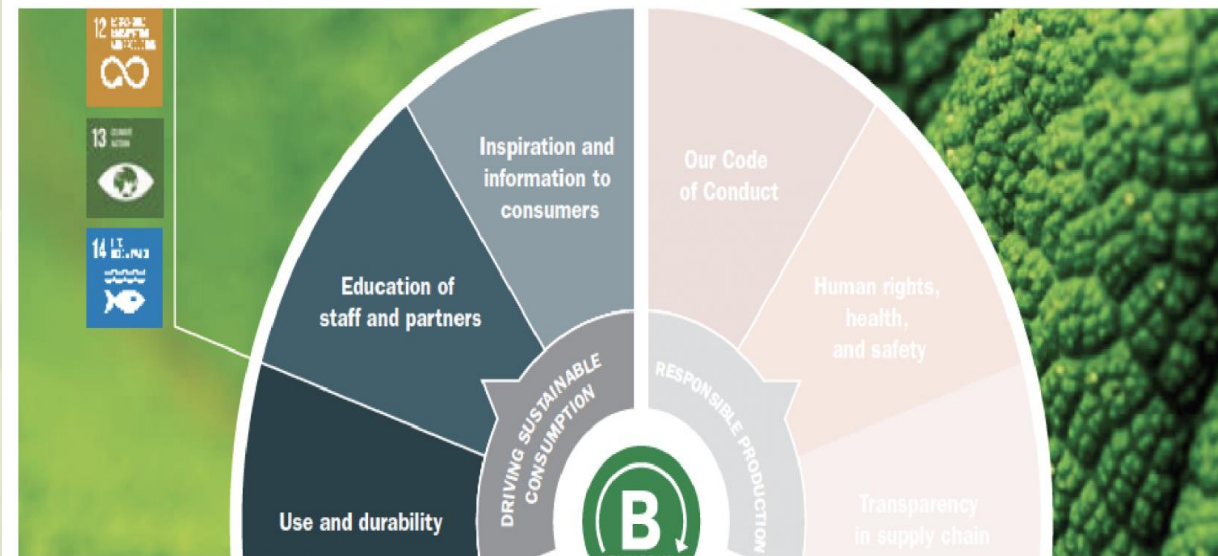
- Public policies are required to create conducive environments and influence the market and economy as a whole, including sustainable public procurement policies and practices to drive sustainable innovation, and smart subsidies to divert investments away from environmentally damaging practices.
- A profound transformation of business practices along global value chains is also required.
- In order to support reform on green fiscal policy, UN Environment has established the Green Fiscal Policy Network in partnership with the International Monetary Fund .
- The Network facilitates knowledge-sharing and dialogue on green fiscal policy reforms, working with a number of associated partners.
- The exceptional situation resulting from COVID-19 has been a good opportunity to reflect on our habits and to see which ones we want to keep and which ones we want to change.



SCP holistic approach



- When we say “**responsible**” to our consumption, we are **adding value to the choice we make every day as consumers** between the different goods and services offered to us by the market.
- It is a **conscious choice of products** that truly meet our needs and, at the same time, respect the environment.
- Therefore, before buying, we should **ask ourselves questions** such as “**do I really need it**”, “am I choosing freely or am I acting compulsively”, “**how long will I use it**”, “**can I borrow it from someone I know**”, “**is it made from recyclable materials**”, “who made the product and how did they make it”, “**who made it and how?**”



❑ Examples to understand **why responsible and sustainable consumption is necessary**:

✓1. We help local commerce-

- **Consuming in a responsible way** also means **supporting local, zero-carbon products**.
- **Shops contribute to the local economy**, which is a very important source in the current context for the recovery of our city. So, one example would be to **buy more from local shops** rather than from large corporations.

✓2. We **take care of the environment**-

- Before we buy, let's think about the **environmental footprint** that goes with it. From the sourcing of the raw materials it is made from, through production and use, to the **management of the final waste**.
- It is very important **to prioritize products with a lower environmental impact** and **reduce the consumption of the most limited natural resources such as water, oil, gas and others**.



✓3. We will protect animals-

- It is no secret that in order to supply the massive demand, the treatment of animals by many corporations has usually been unethical and has been questioned for decades.
- Therefore, by buying eco-friendly, vegan and sustainable clothing, we are also helping to conserve the lives and good treatment of hundreds of species.

✓4. We save money in our homes-

- If we consume responsibly at home, we will buy less and better, and this will lead to financial savings.
- Sustainable consumption also pushes us towards greater energy efficiency and today there are many household appliances and systems along these lines, which make our homes more efficient, reduce the impact on the environment and also generate economic savings.



✓5. We are committed to the three R's-

- Responsible consumption also leads us to recycle, reduce and reuse. A cycle that will also be fundamental for the education of children.
- In this way, we look for alternatives that minimize the exploitation of natural resources, such as, for example, giving a second life to some products through second-hand markets, exchanges or the repair of objects and machinery.

✓6. We have a responsible attitude towards others-

- Responsible consumption values the labour rights of the people who work on the final product that we end up buying.
- If we opt for this type of consumption, we will be defending first and foremost the obligation to guarantee the same rights for all people.



✓7. We value more that which is not material-

- Every day we are exposed to numerous stimuli that invite us to consume and to possess more and more, compulsively and unnecessarily.
- Responsible consumption also implies leaving behind the society of materialism in order to value everything that does not cost money but enriches us enormously and, at the same time, becoming aware that we live better with less.

❑ Food as part of sustainable consumption:

- The issue of food is crucial to our understanding of our responsibility as consumers.
- Studies have shown that the misuse of food is one of the causal factors of the climate emergency we are experiencing.
- In order to respond to the extremely high global demand for food, we resort to abrasive methods and the exploitation of natural resources that lead to deforestation, etc.



❑ How can we promote responsible food?

- ✓ We must plan our shopping well and opt for healthy, local food.
- ✓ It is also a good time to be creative, innovate and create new recipes so as not to waste anything.
- ✓ Responsible consumption is a practice that has to become established in the gastronomic and catering sectors/food industries.
- ✓ Sustainability is our commitment; the new law obliges establishments to reduce their food waste;
- ✓ By making small gestures such as opting for local consumption of products and services, we can generate major positive dynamics for our future in the city and for the future of the planet.
- ✓ In most cases, responsible consumption only involves consciously changing our habits.
- ✓ Each one of us can turn our purchasing power into an important instrument of change towards a more sustainable world.



Reduce		Reuse		Recycle	
To use less		To use again		To make something new with something that has been used	
Using public bike and not riding in a car	Unplugging electronic wires when it's not used	Buying gift bags from a local shop	Using stainless steel instead of plastic forks	Turning empty containers into a bird feeder	Using magazine cuttings to make an art project
Closing the front door when the air is on	Using reusable containers	Donating clothes that are too small	Refilling a water bottle	Using a compost bin	Using pieces of paper to make new paper
Always turn the tap when it's not used	Turning off the lights when you leave a room	Using a reusable grocery store bag	Using non-toxic cleaning products	Using newspaper to make a bed	Placing plastic cups in the recycling bin



• Key business themes addressed by this SDG :

- Sustainable sourcing
- Resource efficiency of products and services
- Materials recycling
- Procurement practices
- Product and service information and labeling
- Examples of key business actions and solutions
 - Implement product portfolio analysis tools to understand **environmental and social footprint of products** within lifestyles as well as production. **Innovation must align products and applications to appropriately address sustainability megatrends.**
 - Develop **innovative business models** such as moving from selling products to selling services, to **retain ownership of the products** and help close the materials loop.

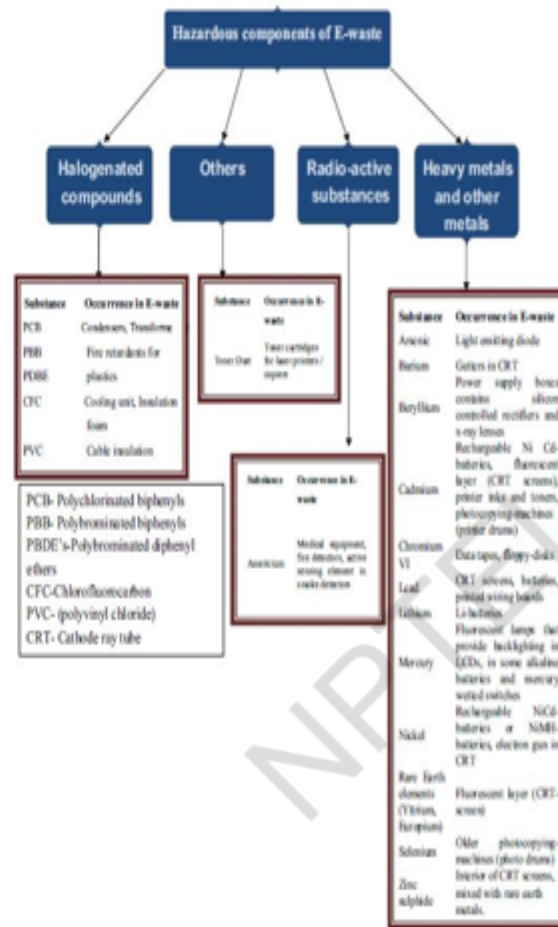


- Enable sustainable consumption by developing **innovative solutions** can reduce energy need in usage and **educate consumers** about these benefits.
- **Reduce manufacturing impacts** by substituting virgin raw materials in products with post-consumer materials through recycling and upcycling.
- Apply **modular design**, so products' constituent parts will be easily separated and either **re-used without further processing, or easily recycled near the point of disposal**.
- Significantly **reduce waste** and **ensure** that any unavoidable waste is utilized to the fullest degree (e.g. **organic waste as fuel or fertilizer**).

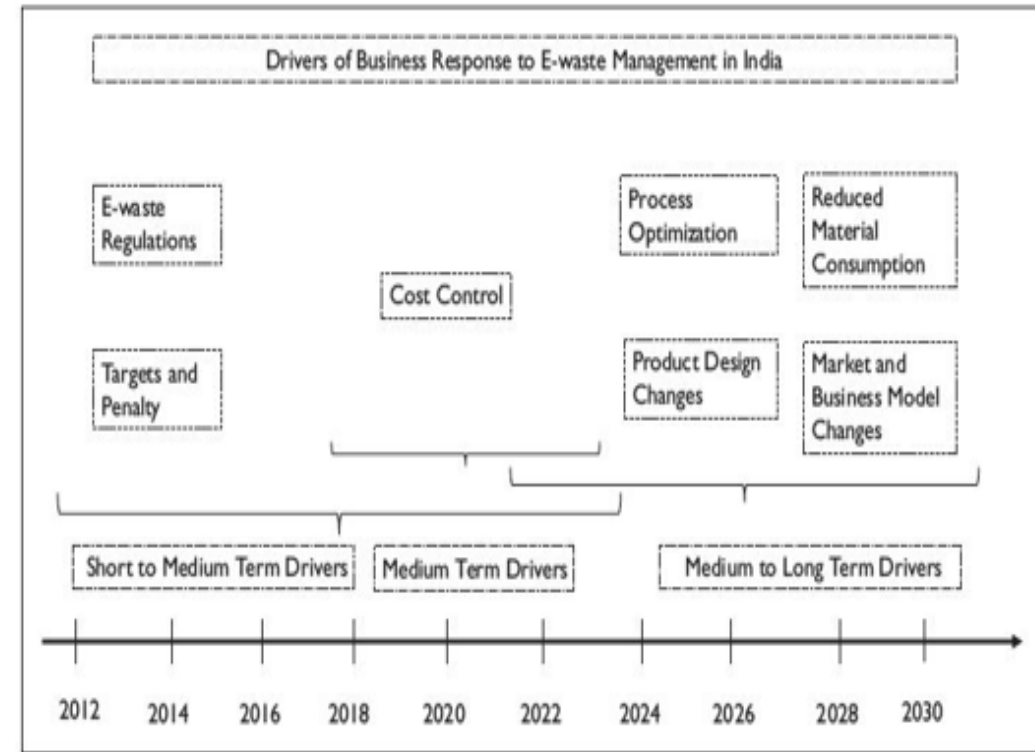


E- WASTE MANAGEMENT

- Refers to all electronic and electrical equipment.
- Refrigerators, mobile phones, computers or even a bulb which are thrown out after the usage.
- Constituents differ considerably on the manufactured products and contain over 1000 different substances that are classified as "hazardous" or "non-hazardous"



CASE STUDY - INDIAN E-WASTE MANAGEMENT SYSTEM



□ FUTURE FOOD TOGETHER : Transforming Food Systems in the Global South

• HOW WE TAKE ACTION

- We work with local experts to develop and implement in-country solutions along the entire value chain, by translating research and expertise into tangible solutions that are actionable and scalable.
- We build on tailored bottom-up solutions instead of simply following general frameworks in a top-down manner.
- We empower knowledge exchange and participation to facilitate insight sharing between countries in the Global South.
- *Future Food Together* rallies stakeholders at all levels, facilitating engagement and cooperation within international frameworks, as well as with governments, businesses and consumers.

For Governments

For policymakers, this means that integrated, coherent and cross-sectoral policies must be developed and implemented with urgency.

FOR
GOVERNMENTS

For Businesses

For companies, this means aligning their business strategies with both the Sustainable Development Goals and planetary boundaries.

FOR
BUSINESSES

For Consumers

For each of us, this means making more informed choices and advocating for healthier and more sustainable lifestyles.

FOR
CONSUMERS



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NPTEL



ESD FOR GLOBAL PARTNERSHIP

Lecture: 58

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□ Goal 17. Strengthen the means of implementation and revitalize the global partnership for sustainable development

Finance

- 17.1 Strengthen domestic resource mobilization, including through international support to developing countries, to improve domestic capacity for tax and other revenue collection
- 17.2 Developed countries to implement fully their official development assistance commitments, including the commitment by many developed countries to achieve the target of 0.7 per cent of

Technology

- 17.6 Enhance North-South, South-South and triangular regional and international cooperation on and access to science, technology and innovation and enhance knowledge sharing on mutually agreed terms, including through improved coordination among existing mechanisms, in particular at the United Nations level, and through a global technology facilitation mechanism
- 17.7 Promote the development, transfer, dissemination and diffusion of environmentally sound technologies to developing countries on favourable terms, including on concessional and preferential terms, as mutually agreed
- 17.8 Fully operationalize the technology bank and science, technology and innovation capacity-building mechanism for least developed countries by 2017 and enhance the use of enabling technology, in particular information and communications technology

Capacity-building

- 17.9 Enhance international support for implementing effective and targeted capacity-building in developing countries to support national plans to implement all the sustainable development goals, including through North-South, South-South and triangular cooperation

Trade

- 17.10 Promote a universal, rules-based, open, non-discriminatory and equitable multilateral trading system under the World Trade Organization, including through the conclusion of negotiations under its Doha Development Agenda
- 17.11 Significantly increase the exports of developing countries, in particular with a view to doubling the least developed countries' share of global exports by 2020



SDG 17: Targets and Indicators

- The UN has defined **19 Targets and 25 Indicators for SDG 17**. Targets specify the goals and Indicators represent the metrics by which the world aims to track whether these Targets are achieved.
- SDG 17 is a vision for strengthening and streamlining cooperation between nation-states, both developed and developing, using the SDGs as a shared framework and a shared vision for defining that collaborative way forward.
- It seeks to promote international trade, and help developing countries increase their exports to ensure a universal rules-based and equitable trading system that is fair, open and beneficial to all.

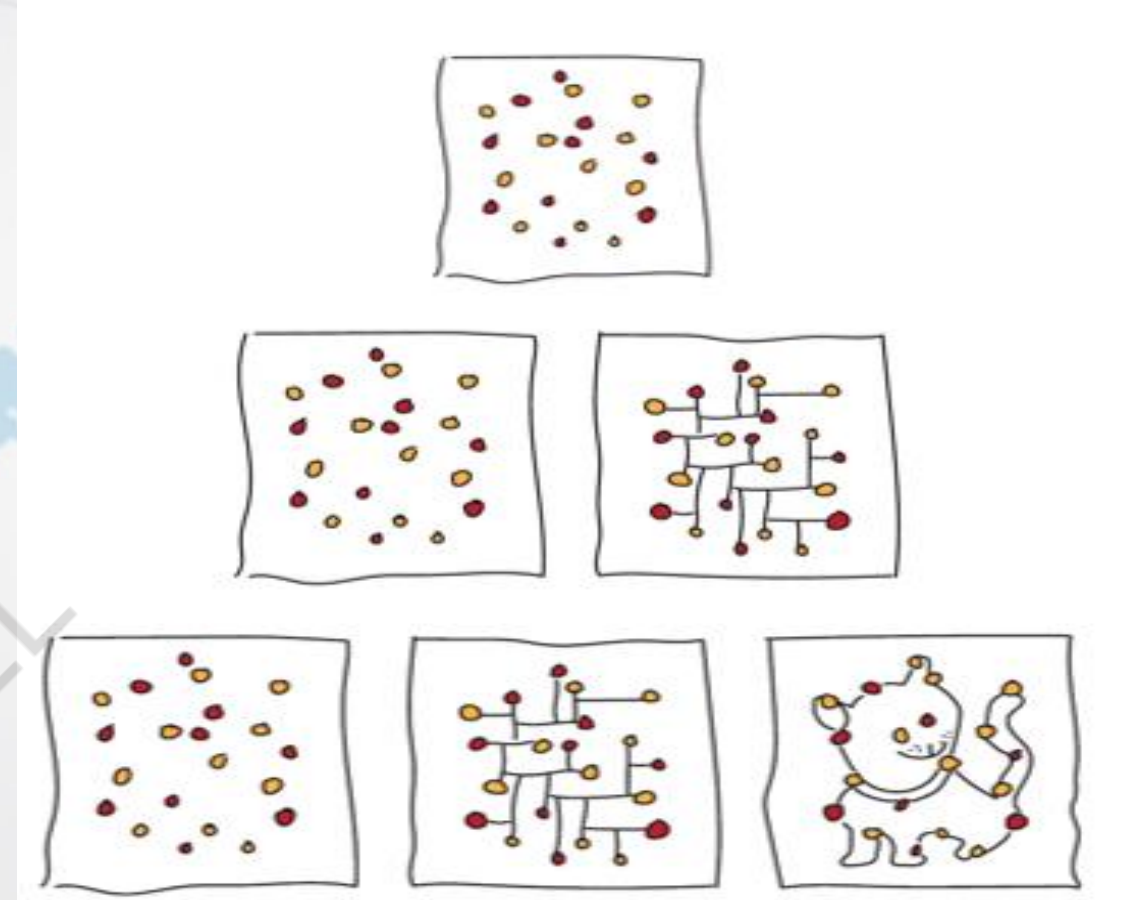


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❑ Partnership to Maximize Impact

- ❑ *"It is the long history of humankind that those who learned to collaborate and improvise most effectively have prevailed."- Charles Darwin*
- ❑ *"A successful sustainable development agenda requires partnerships between governments, the private sector and civil society. These inclusive partnerships built upon principles and values, a shared vision, and shared goals that place people and the planet at the center, are needed at the global, regional, national and local level."- UN*
- All of the ideas, people, technologies, institutions and resources that are required to achieve the SDGs are already available, and the task is **how do we engage them and combine them in new and transformational ways?**



❑ The 2030 Agenda Partnership Accelerator

❑ It is a collaborative initiative of United Nations Department of Economic and Social Affairs (UN DESA) and The Partnering Initiative (TPI), in collaboration with United Nations Office for Partnerships, UN Global Compact, and the UN Development Coordination Office.

➤ Aims:

- ✓ Raise understanding and build the partnering skills and competencies
- ✓ Support the organizational change required for our institutions to become 'fit for partnering'
- ✓ Draw out good practice and support the development of efficient SDG partnership platforms

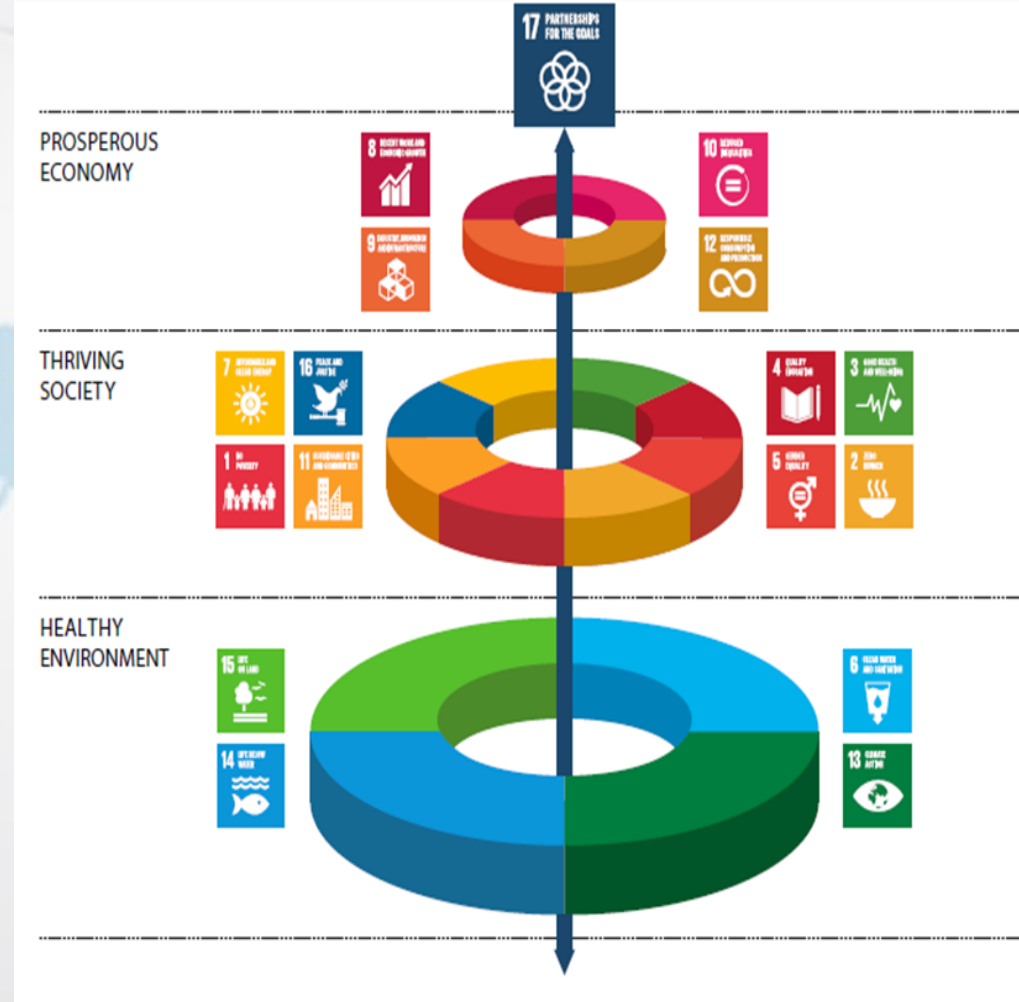
(Source: sustainabledevelopment.un.org)



2030 Agenda and Essential Role of Partnerships

(Source: SDG Partnership Guidebook)

- Built on an understanding of the **interconnectedness** of the prosperity of business, society and the environment.
- All **societal sectors** (including business) recognized as key development actors and part of the solution
- Essential **need for collaboration** across societal sectors
- **Need for holistic approaches** across issues and geographies to tackle systemic challenges
- Longer term investment required for **transformational change**
- **Need for innovative approaches** with greater long term potential to tackle complexity but greater risk of failure
- **Emergent planning** based on the coalescing of interests and local resources around particular issues.



❑ Understanding Different Stakeholders

➤ GOVERNMENTS

- Ensure democratic representation of the people
- Deliver national defense
- Maintain law and order
- Provide a stable, regulated environment for trade
- Collect taxes
- Provide public services
- Provide public infrastructure

➤ BUSINESS

- To produce and distribute goods and services (directly, or as part of a value chain) to satisfy a public need or demand;
- To make a financial profit for investors / owners.



➤ CIVIL SOCIETIES

- (Particularly international NGOs): Access to **international knowledge and resources**;
- Technical knowledge / **delivery capacity**;
- **Deep knowledge** of, and reach and access to, communities and people;
- Legitimacy / **social capital** / influence (can be particularly strong in faith-based organizations);
- Ability to organize and **engage people** (e.g. around advocacy)

➤ UNITED NATIONS

- To support the government in **building and strengthening national capacities** and delivering the national development agenda:
- Promoting sustainable development
- Delivering **humanitarian aid**
- Upholding **human rights**
- Upholding **international law** and maintaining **peace and security**



➤ FOUNDATIONS

- Providing funding and/or technical assistance and/or running programs using its own sources of funding to deliver the foundation's goals.

➤ DEVELOPMENT COOPERATIONS (DONORS)

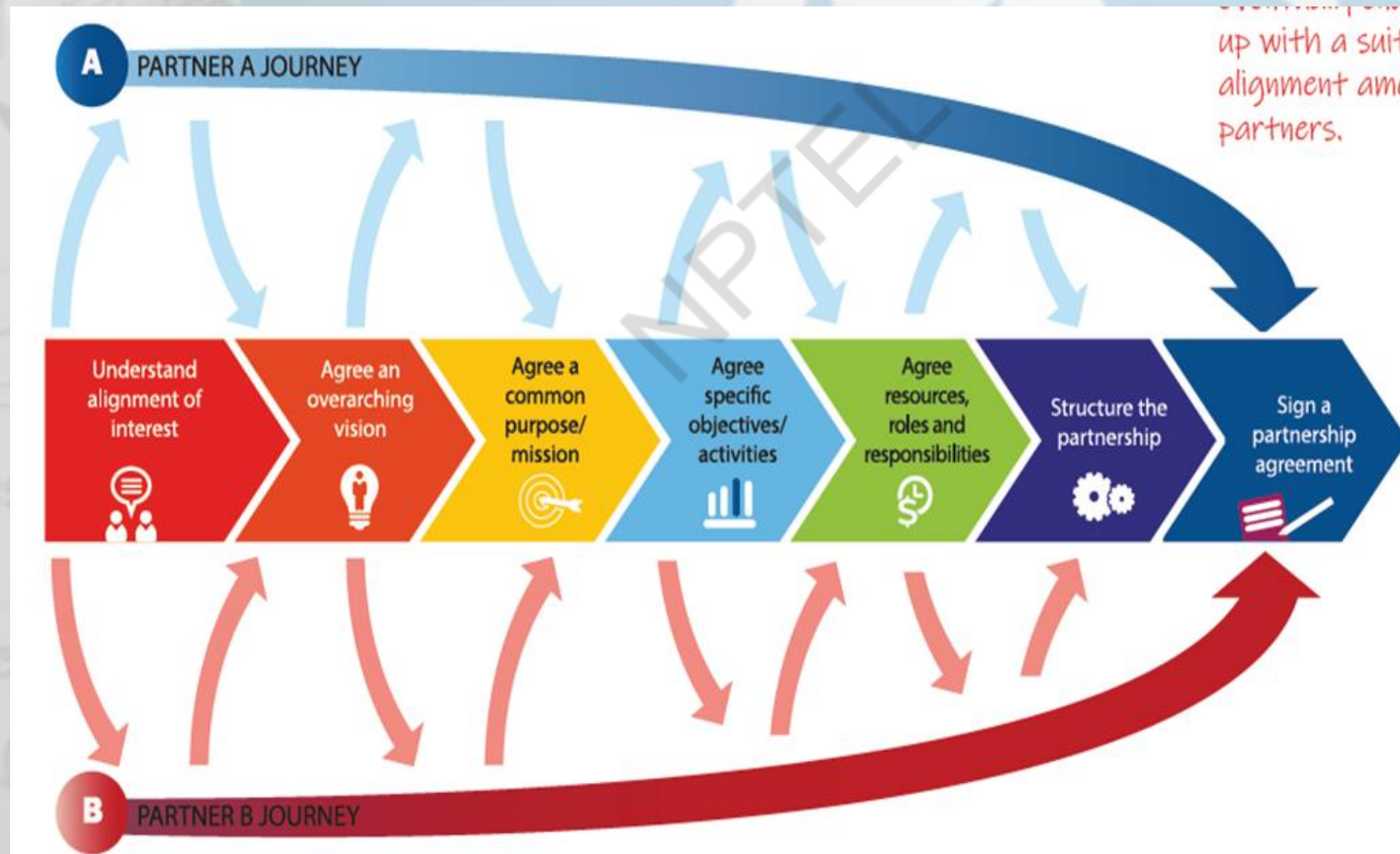
- Funding
- Political connections and influence
- Technical assistance

➤ OTHER STAKEHOLDERS

- Academia, Media, Tribes, Parliamentarians, Trade Unions



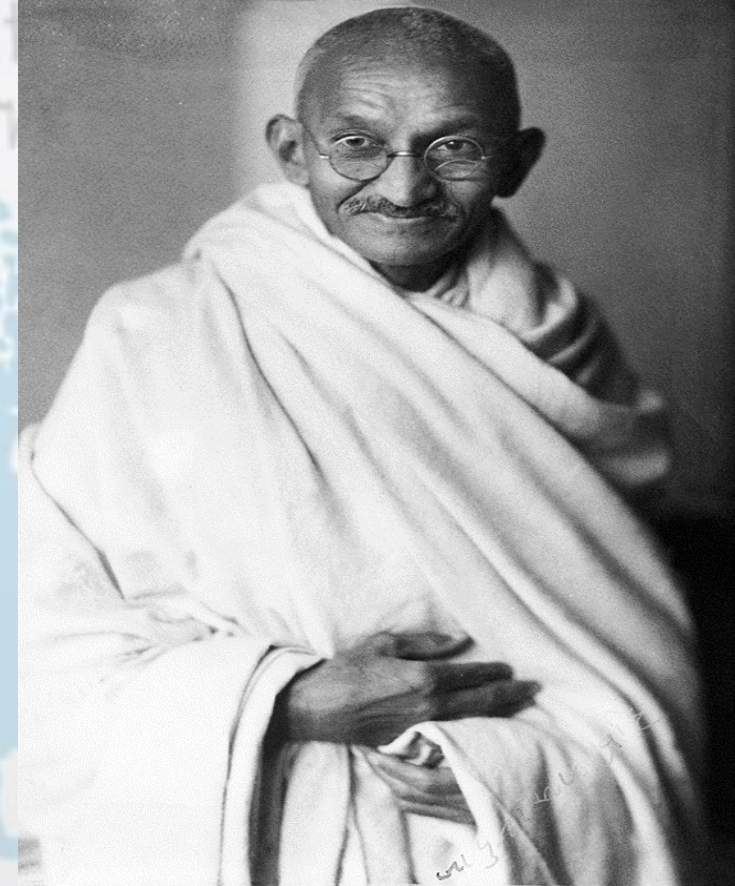
- **Partnership Journey to Action**
- Post-agreement journey
- Innovative thinking and planning
- Implementation Challenges become Evident
- Settling in of Management Systems
- Performance



• The Ideals and Relevance of Mahatma Gandhi's Ideals

• Non-violence and Partnerships

- Gandhiji was perhaps the most vociferous leader the human race has seen on the matter of **non-violence** and the **freedom movement of India** was the only example in history of the world of a country getting independence without taking up arms.
- Non-violence is a gospel in today's world and if the gospel is followed at the **political scenario of the world** accompanied for less greed and hunger for power, the world would be closer to what the SDGs envision.
- Just like differences and conflicts at micro-level the differences and **conflicts at international levels** can be sought out with understanding, cooperation to good, non-cooperation to evil and arbitration.



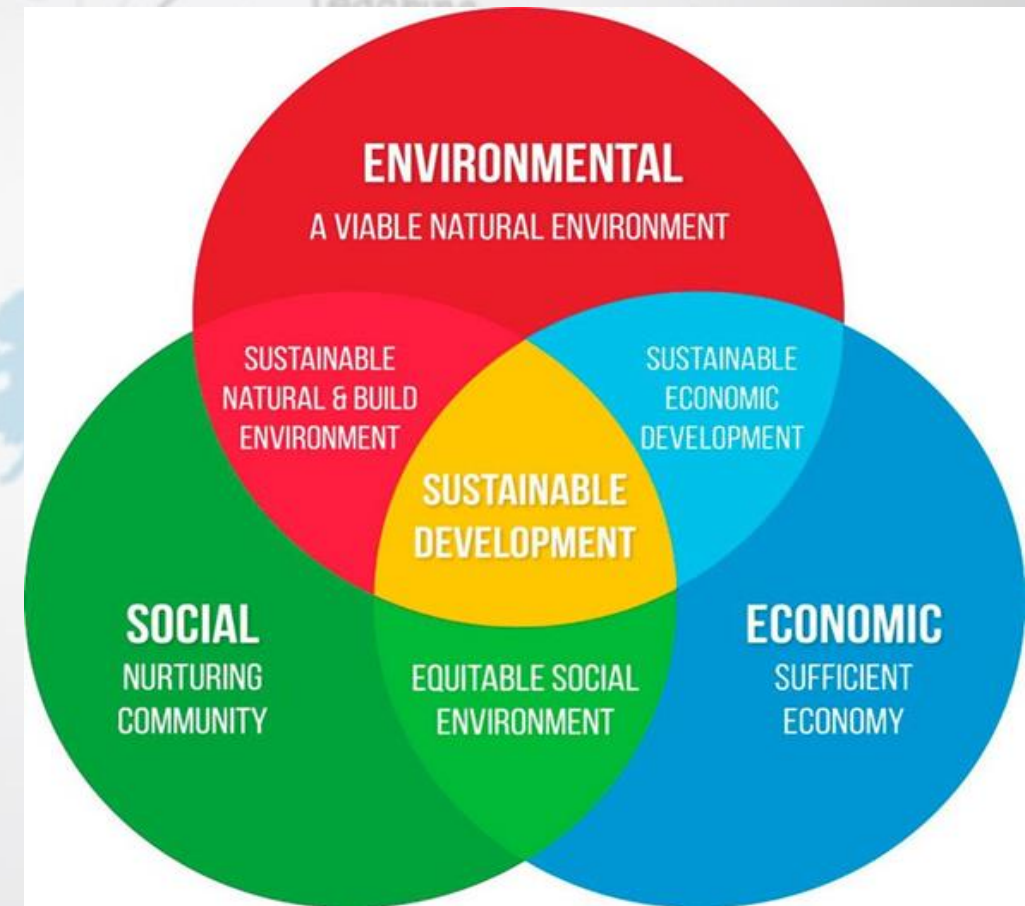
❑ Idea of Trusteeship and Corporate Social Responsibility

- The essence of commerce, for him, is that the wealth generation should not be at the cost of individual, the generated **wealth should be used for the betterment of the humanity** and not for personal indulgence.
- “**I want to save time and labour not for a fraction of mankind but for all. I want the concentration of wealth not in the hands of a few to ride on the back of millions but in the hands of all.**” (Gandhi, 8)
- That is how the idea of trusteeship was born and many industrialists gave their wealth **for the betterment of society**.
- This idea was inspired by Gandhiji more than 100 years ago and business tycoons gave wealth for uplifting of mass and society.
- This concept took the name of ‘**Corporate Social Responsibility**’ –CSR



- **SD Pillars and Gandhian Ideals**

- The 17 goals and its 169 targets of SDGs, which seek to address the issues and concerns pertaining to the **three pillars of sustainable development- social equity, economic development and environmental justice-** are connected to each other, **Gandhi's ideas of ethical living and value-based approach hold good for SDGs.**
- **His eternal call for 'truth' (Satya) and 'non-violence' (ahimsa) is as relevant today** as it was long ago for integrated development of any society.
- His **techniques of 'satyagraha' (non-violent protest) carry the means to combat many a social and environmental threats**, be it climate change, natural resource depletion.





ESD FOR GLOBAL PARTNERSHIP

Lecture: 59

Prof. Atasi Mohanty
Rekhi Centre of Excellence for the Science of Happiness

• Gandhian Views and SDGs

- The '**Khadi Movement**' initiated and spearheaded by him is the preeminent prototype of **small and cottage industries**. **Strengthening village economy**, and thus, reducing excessive urbanization can make human settlements safe, resilient and sustainable (**SDG-11**).
- Gandhi has all the time opposed unplanned and irresponsible industrialization, and has **recommended for sustainable industrialization and local innovation** (**SDG-9**) through use of indigenous and miniature technologies.
- Gandhi has persuaded the communities for **management of water, sanitation and hygiene in a sustainable way** (**SDG-6**).
- As a visionary and sensitive environmentalist Gandhi was probably aware of and has cautioned against, without naming, the **environmental crisis** the world might face like **climate change** and has offered means to combat it (**SDG-13**).
- He has appealed for **protection of environment and ecosystems** (**SDG- 14 & 15**) by rational **use of natural resources**, promotion of water harvesting, use of **organic manure**, ending cruelty to animals, and large-scale afforestation.



➤ Indian Involvement

- India has joined **International Solar Alliance**, India has committed to Paris deal on Climate Change, etc.
- 2019: India became a **founding member of the European commission's International Platform on Sustainable Finance**
- NITI has set no quantitative targets for this Goal because MOSPI is unable to frame statistical methods to capture it. (e.g., **SDG Target 17.15**)
- In **2021's SDG Index Report**, NITI provided some qualitative assessments, e.g. we need to increase the co-operation at the global level, Corona has created new challenges, etc.
- **CoP25 MADRID, Spain in 2019**
- **UNFCCC → 25th Conference of the Parties (COP 25)**
- India hosted '**India Pavilion**' at **COP-25** with the theme '150 years of celebrating the Mahatma' to highlight **sustainable living**.



➤ COVID-19 Impact

- Covid-19 has exacerbated the **problem of resource mobilization**, putting India in jeopardy at a time when western economies are withdrawing from different international agencies, reducing back, and tapering financial support.
- There is a **need to strengthen ties with neighboring nations** so that India may better mobilize resources for its economy.
- The **economies of Southeast Asia** provide us with a large market for our completed products, with **exports playing a significant role**.
- **Co-operation among countries** is the need of time, be it in vaccine supply efficiently to developing and underdeveloped countries or following the NDCs of Paris in Letter and Spirit.



➤ UNESCO ON GANDHI

- **UNESCO is the flag-bearer of SDGs**
- UNESCO constitution which states, “that a peace based exclusively upon the political and economic arrangements of governments would not be a peace which could secure the unanimous, lasting and sincere support of the people of the world, and that the peace must therefore be founded if it is not to fail, upon the intellectual and moral solidarity of mankind”
- **Culture of Peace promotion** is one of the primary objective of UNESCO



United Nations
Educational, Scientific and
Cultural Organization



• GLOBAL CITIZENSHIP

- interpreted as 'citizenship beyond borders', or 'citizenship beyond the nation-state'
- Global citizenship does not entail a legal status
- future society will be heavily dependent on the international cooperation on economic and trade fronts
- focus on global issues
- cognitive skills as well as non-cognitive skills

• SOLUTION



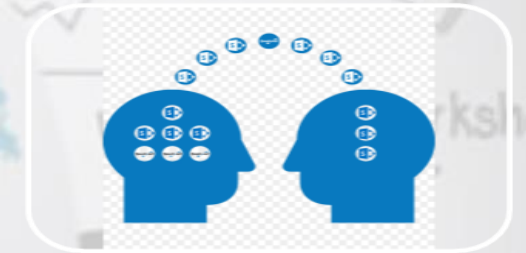
Global
Citizenship



EIU



International
Cooperation



Knowledge
sharing



➤ EDUCATION FOR INTERNATIONAL UNDERSTANDING (EIU)

- promotes international goodwill through education
- educate the minds of young people psychologically and intellectually
- Choosing what is wrong and what is right
- child should be told from the very early age that the whole world is one family
- Promotes 'global culture'



➤ SHARING OF KNOWLEDGE AND EXPERIENCES THROUGH INTERNATIONAL COOPERATION

- ICT have enabled **people to connect and interact** with others around the globe anywhere, anytime
- **sharing of information and experiences** at regional and global levels
- participate in **inter-governmental cooperation and partnerships** for global benefit
- **periodic discussions** can continually renew interests and reconstruct the objectives of **global citizenship education**



➤ ACCESS TO INTERNET

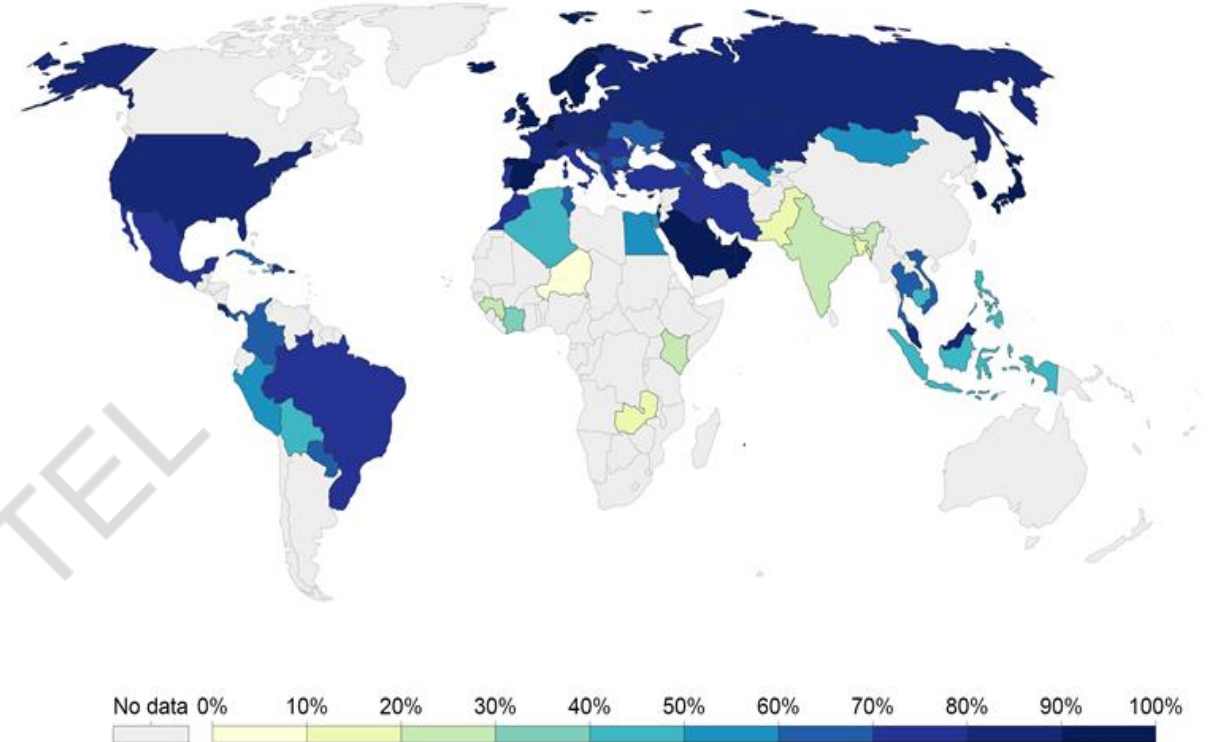
➤ SDG INDICATOR 17.8.1

- proportion of individuals using the Internet.
- India 20.08% of the population have access to internet (2018 data)



Share of the population using the Internet, 2019

All individuals who have used the Internet in the last 3 months are counted as Internet users. The Internet can be used via a computer, mobile phone, personal digital assistant, games machine, digital TV etc.



Source: International Telecommunication Union (via World Bank)

OurWorldInData.org/technology-adoption/ • CC BY



➤ FIXED BROADBAND CONNECTION

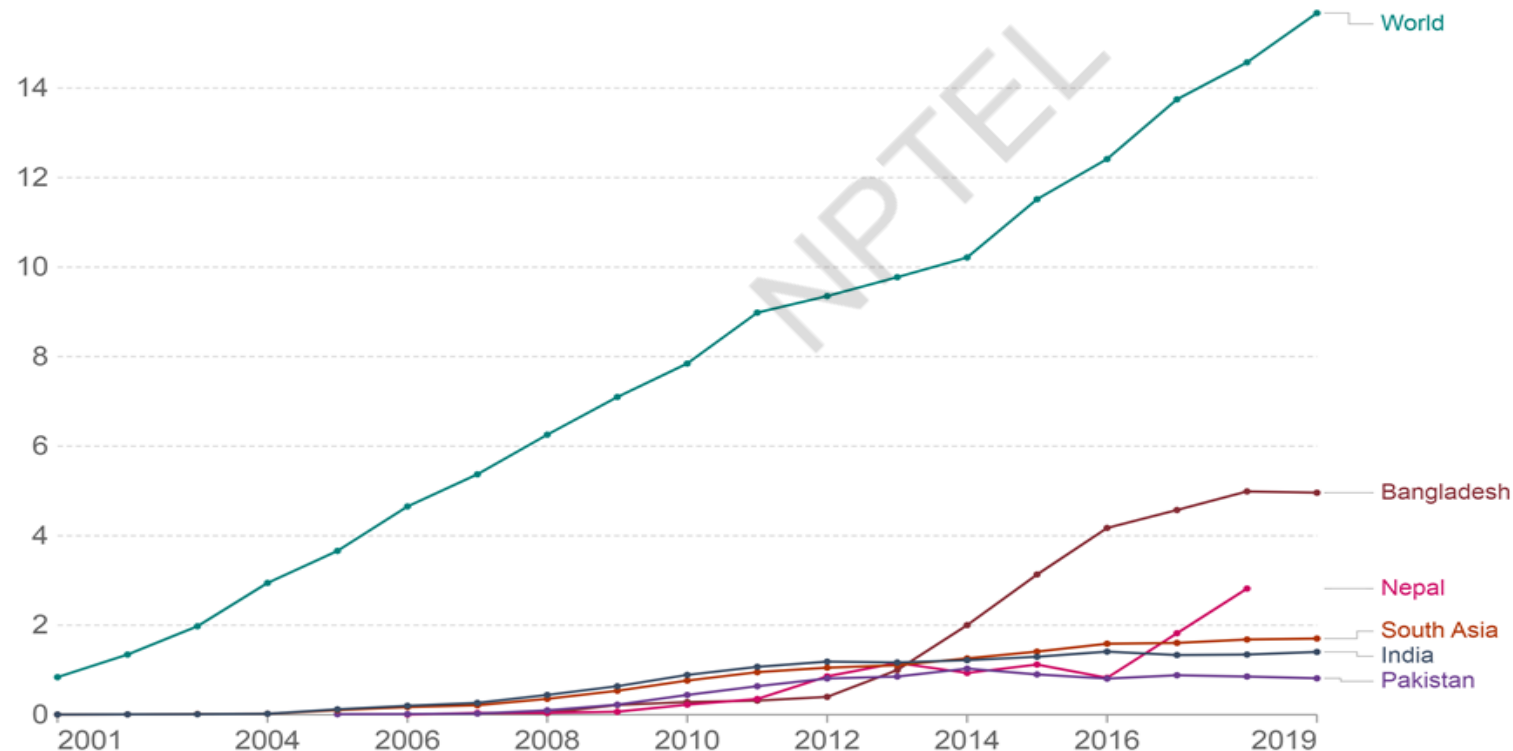
➤ SDG INDICATOR 17.6.1

- 1.4/100 have fixed broadband connections in India
- According to 2019 data

Fixed broadband subscriptions (per 100 people), 2001 to 2019

Fixed broadband subscriptions refers to fixed subscriptions to high-speed access to the public Internet (a TCP/IP connection), at downstream speeds equal to, or greater than, 256 kbit/s.

Our World
in Data



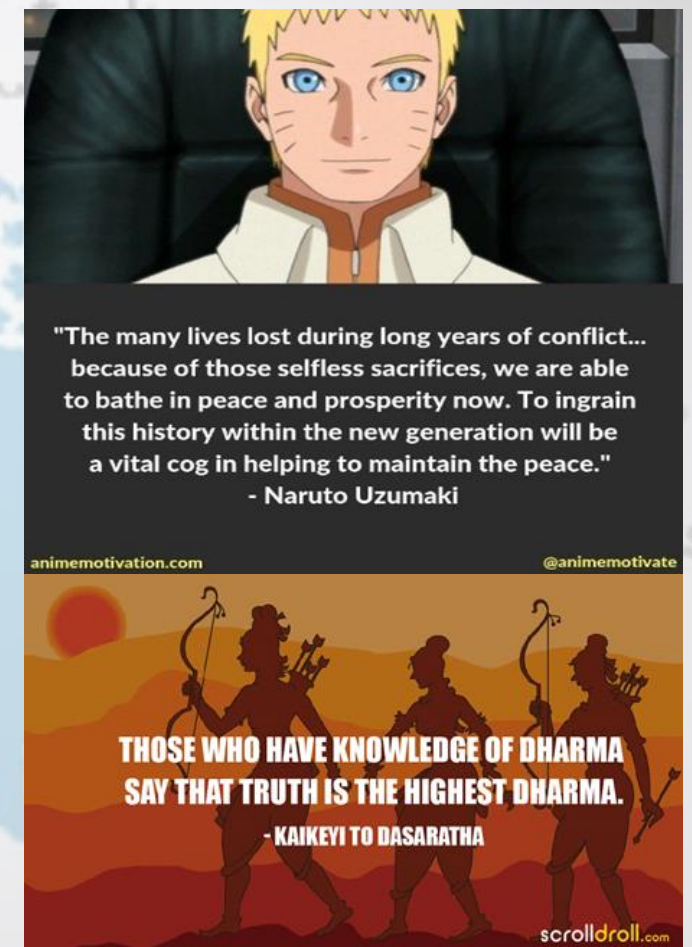
Source: International Telecommunication Union (via World Bank)

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➤ HOW TO IMPLEMENT

- Gandhian principles should be inculcated in the ethos of the present and upcoming generations
- Reflect the goal of global citizenship education in countries' curricula
- A transformative, dynamic and flexible pedagogy has to be formulated
- Learning should be accessible to all the stakeholders at all time anywhere in the globe
- Global campaigns and advertisements should be committed in order to promote peace and non-violence.
- Encourage youths to take on complex challenges
- research and peace-dialogues to resolve conflicts



Milestones in the Global Journey from

The Millennium Development Goals (MDGs) are agreed.

MDG 8 on developing a global partnership for development recognizes the importance of inclusive international efforts.

Millennium Summit of the United Nations, New York

2000

Funding targets are set to achieve the MDGs.

Donor countries should allocate 0.7% of their gross national income to development assistance.

Financing for Development Conference, Monterrey

2002

Donors are asked to coordinate their activities to reduce transaction costs for recipients of development assistance.

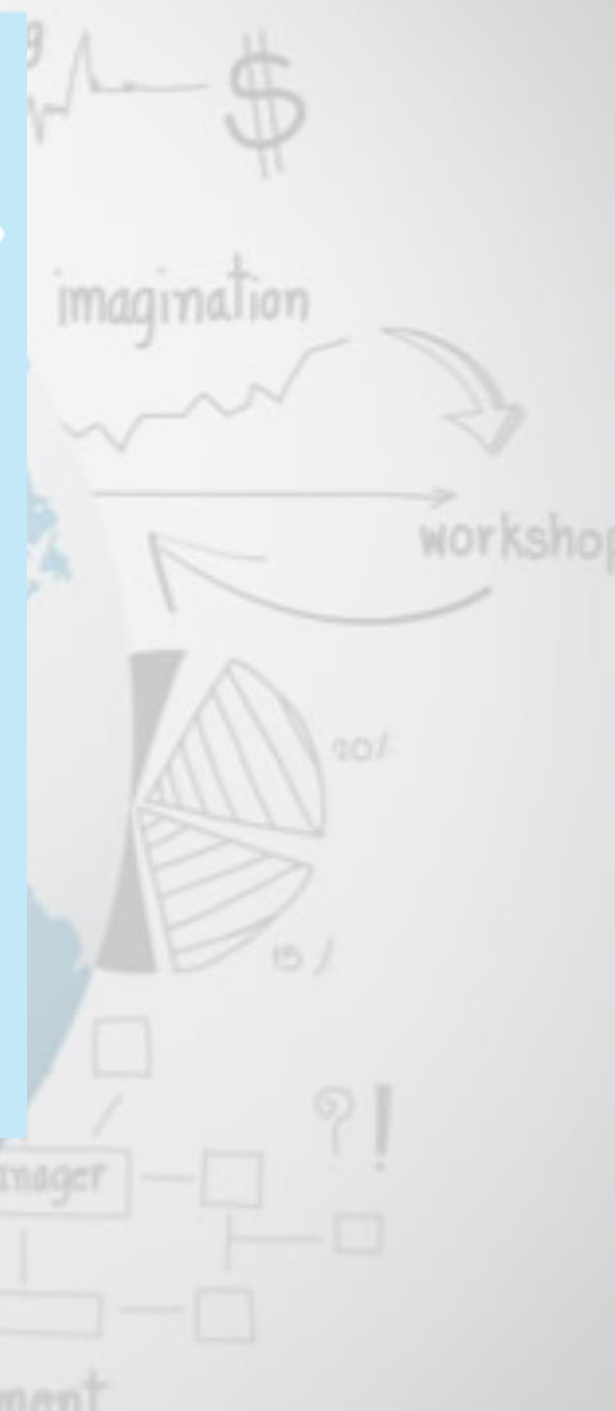
High-Level Forum on Harmonization (HLF-1), Rome

2003

The Paris Declaration establishes five principles for aid effectiveness: country ownership of the development agenda, donor alignment with country priorities and systems, harmonization of donor policies, managing for development results, and mutual accountability.

Second High-Level Forum on Aid Effectiveness (HLF-2), Paris

2005



Aid Effectiveness to Development Effectiveness

The Accra Agenda for Action upholds the Paris Principles, emphasizing the role of emerging economies, civil society organizations, and parliamentarians in development.

Third High-Level Forum on Aid Effectiveness (HLF-3), Accra

2008

The Bogota Statement commits partners engaged in South-South cooperation to deepen knowledge exchange and mutual learning.

The Dili Declaration proposes to counter conflict and fragility.

The Istanbul Principles provide guidance for the development work and practices of civil society organizations.

Bogota, Dili, and Istanbul

2010

The Global Partnership for Effective Development Cooperation embodies the shift from aid effectiveness to effective development cooperation.

Fourth High-Level Forum on Aid Effectiveness (HLF-4), Busan

2011

The Busan principles for effective development cooperation are reaffirmed.

Action is agreed on the priority themes of domestic resources mobilization, middle-income countries, business as a partner, South-South cooperation, triangular cooperation, and knowledge sharing.

First High-Level Meeting of the Global Partnership for Effective Development Cooperation, Mexico

2014

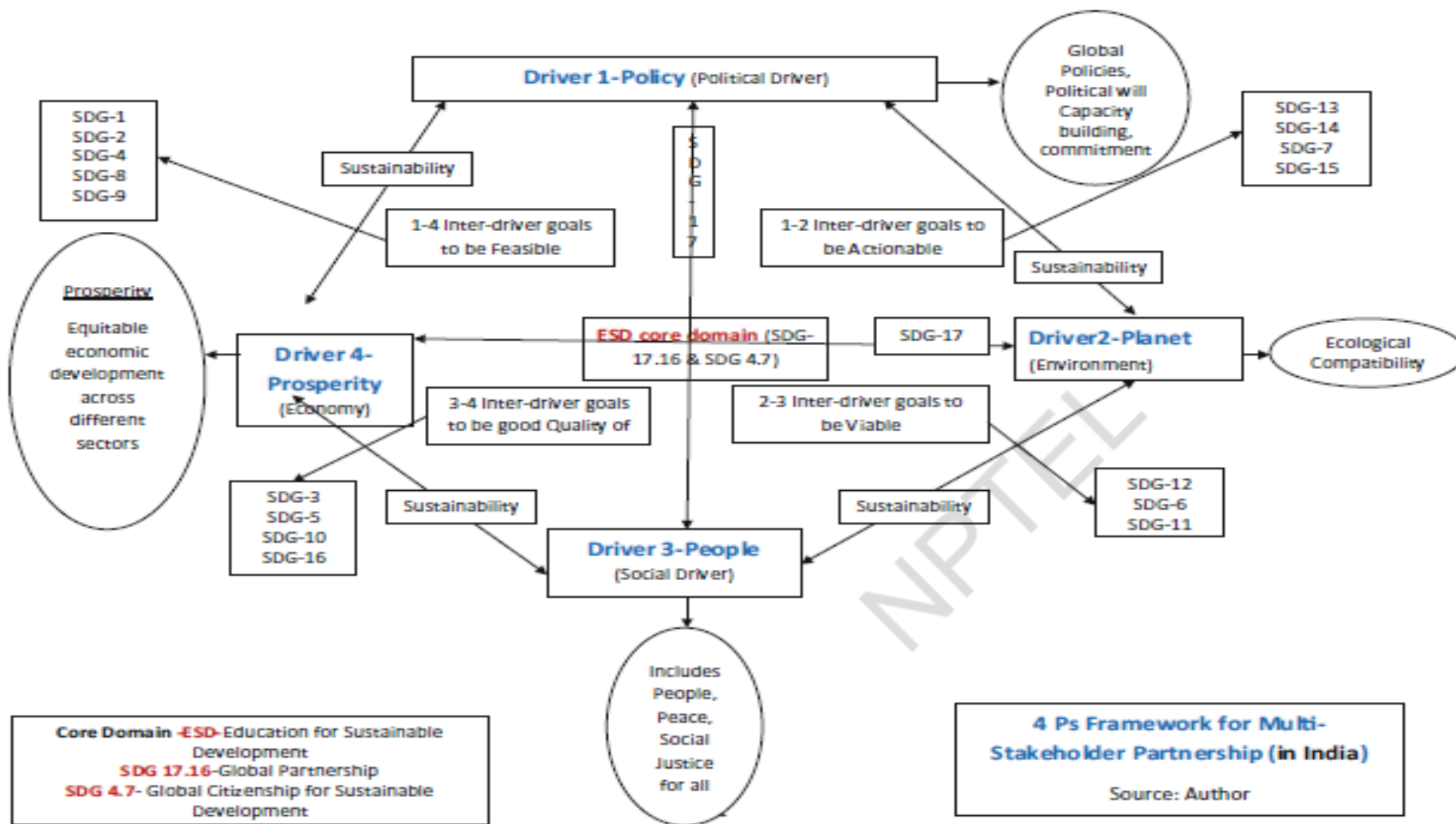


**First High-Level Meeting of the Global Partnership for Effective Development Cooperation
15-16 April 2014, Mexico City**



More than 1,500 development partners worldwide gathered to review global progress in making development cooperation more effective, agree on actions to boost progress, and discuss the post-2015 development agenda.





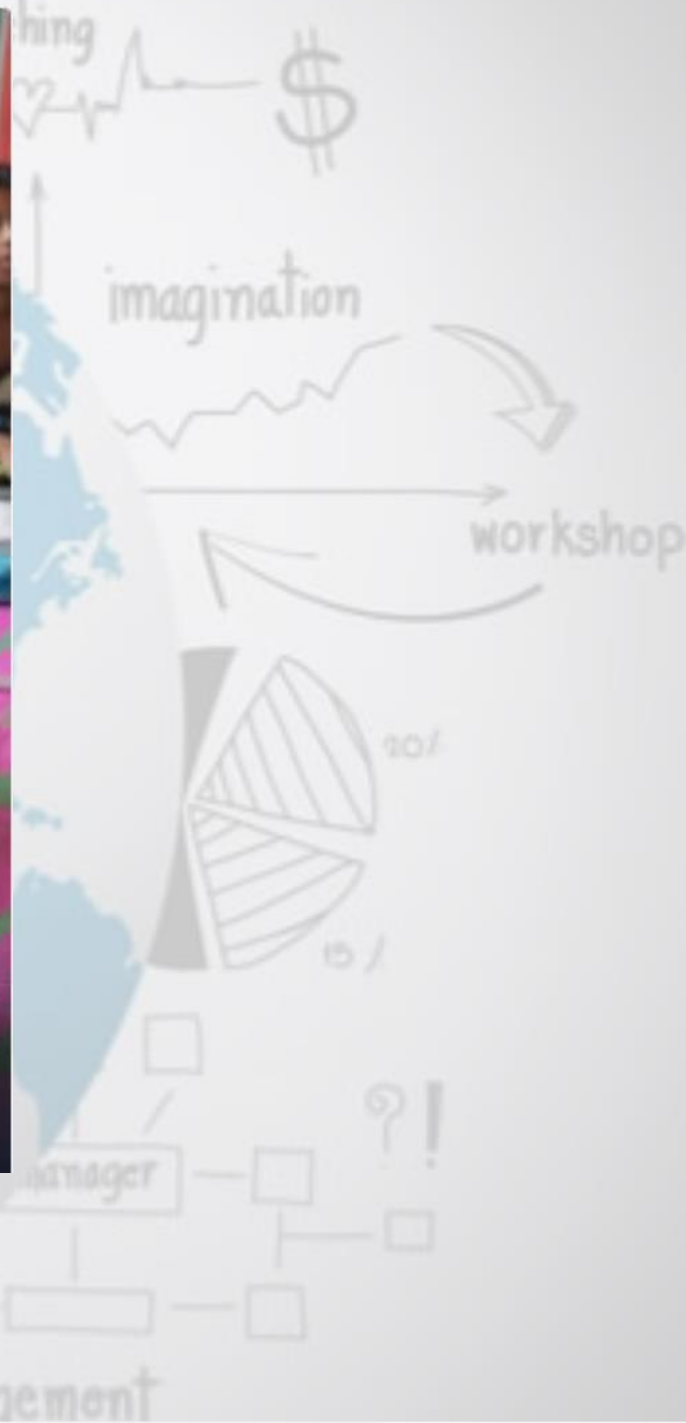
Education for Sustainable Development: A Framework for Global Partnership and Sustainability in India,
Fig. 1 4 Ps framework for Multi-stakeholder Partnership (in India). (Source: Author)





February 08, 2022

Statement by outgoing Chief Executive Officer of GPE, Alice P. Albright





January 24, 2022

Statement by GPE Board Chair on the International Day of Education

On the international day of education, GPE Board Chair recalls the urgency for the global community to do more for Education, before COVID creates a permanent fracture in the already inexcusable gap between...



January 24, 2022

Japan pledges over US\$8 million to the Global Partnership for Education

The funds will support learning continuity for the most vulnerable children whose lives have been deeply affected by ongoing conflict, violence and food insecurity.



January 10, 2022

Denmark and the LEGO Foundation pledge over US\$15 million in matched contributions to GPE's Girls' Education Accelerator

The Global Partnership for Education (GPE) welcomed 100 million Danish Kroner (approximately US\$15 million) in new pledges



What Does Impact Mean to Global Partnerships?

All of our funds' impact-first investing is aimed at empowering low-income and marginalized people to earn a living, provide the basic necessities of life for their families, and improve their lives. We aim to deliver the highest possible impact...

Broader opportunity – support sustainable solutions across all facets of poverty: livelihoods, education, energy, health, housing, water and sanitation

Deeper inclusion – prioritize investments where opportunity is expanded to include poorer and marginalized people, especially women and the rural poor

Serving millions – invest in sustainable, and therefore scalable solutions, that positively impact the lives of millions of people

Improving lives – focus investments on solutions that empower people to earn a living and provide for the basic necessities of life

...while aiming to preserve capital for all investors.

<https://globalpartnerships.org/about-us/>



Why Are Global Alliances Important?

It is effective to establish strategic global businesses as a means of entering new international markets. There is the possibility for Partners to provide proven products, distribution networks, and marketing support, as well as understand specific markets, allowing them to ensure the products go market faster and are more likely to be sold.

What Is A Global Strategic Alliance?

Global Strategic Alliances are no acquisitions and are commonly set up between two or more groups in their home country over a long period of time. Their purpose is to share ownership of a new venture and establish competitive advantages in their respective countries.

What Are The Three Types Of Strategic Partnerships?

As we mentioned earlier, strategic alliances can be classified into three categories, including joint ventures, equity strategic alliances, and non-equity strategic alliances.

What Are The 5 Components Of A Strategic Relationship?

- An important part of the company's strategy.
- In addition to having a competitive advantage, one's core competency is also important.
- Taking measures to block a competitive threat...
- Our future options are...
- A risk mitigation technique is used to reduce the risk.



What Are The 4 Types Of Partnership?

- A general partnership is one of the most common forms of business cooperation...
- “Limited Partnership.” A Limited Partnership is like a family business but with state-approval....
- It is a limited liability partnership...
- The Limited Liability Partnerships consist of limited liability shares.

What Are Global Strategic Partnerships With Examples?

Strategic alliances are extremely common examples of Starbucks and Barnes & Noble striking an agreement. Coffee is brewed by Starbucks. There is a Barnes & Noble stock of books. In both cases, the companies do what they do best, but at the same time they share space costs to gain each other's benefit.

[The U.S. and India: A Comprehensive Global Strategic Partnership - YouTube](#)



- The United Nations Interagency Task Team on Science, Technology and Innovation for the SDGs will promote coordination, coherence, and cooperation within the UN System on STI related matters, enhancing synergy and efficiency, in particular to enhance capacity-building initiatives. The Task Team will draw on existing resources and will work with 10 representatives from the civil society, private sector, the scientific community, to prepare the meetings of the Multistakeholder Forum on Science, Technology and Innovation for the SDGs, as well as in the development and operationalization of the on-line platform, including preparing proposals for the modalities for the Forum and the on-line platform. The 10 representatives will be appointed by the Secretary General, for periods of two years. The Task Team will be open to the participation of all UN agencies, funds and programmes, and ECOSOC functional commissions and it will initially be composed by the entities that currently integrate the informal working group on technology facilitation, namely: UN Department of Economic and Social Affairs, United Nations Environment Programme, UNIDO, United Nations Educational Scientific and Cultural Organization, UNCTAD, International Telecommunication Union, WIPO and the World Bank.
- The on-line platform will be used to establish a comprehensive mapping of, and serve as a gateway for, information on existing STI initiatives, mechanisms and programmes, within and beyond the UN. The on-line platform will facilitate access to information, knowledge and experience, as well as best practices and lessons learned, on STI facilitation initiatives and policies.



Teaching Sustainability in Higher Education: Pedagogical styles that make a difference

The challenge of teaching sustainable development in higher education can mean that students—as future citizens—are left without insight, commitment, or a sense of their position regarding meaningful beliefs and actions related to sustainability. A paradox arises when educators approach a sustainability curriculum that has the potential to transform students' thinking and actions, with a reductive and non-substantive pedagogy. This paper uses an epistemological and pedagogical analysis of relevant literature to redefine, clarify,

and provide a more systematic and holistic understanding of a transformative pedagogy required for learning. The central thesis juxtaposes three sustainability curricular positions with three pedagogical models that vary decidedly in their emphasis on the prerogative of the learner's prior knowledge and beliefs, the engagement of the learner, and the potential for critical thinking and transformative learning. It is found that a transformative pedagogy overcomes and eliminates the paradox, helping societies become more sustainable

https://www.researchgate.net/publication/322553955_Teaching_Sustainability_in_Higher_Education_Pedagogical_styles_that_make_a_difference



ICT and Green Sustainability Research and Teaching

The centrality of Information and Communications Technology (ICT) in modern life means that ICT plays a part in many aspects of human activity. The search for ways to mitigate the effects of climate change is no exception, and the many potential roles which ICT can play, including the ability to enable data gathering for monitoring purposes; to deliver alternative ways of working; to change supply chains and to control transport and energy supply are well understood. Indeed, examples like these are widely used in predictions of our future. However, ICT has a dual role: as an enabler of change it can support changes to societal behaviour which can reduce the environmental impact of life, housing, work and leisure activities, but as a significant user of resources, it is a contributor to the adverse effects. In order to fully reap the benefits of ICT's contribution while limiting the resource requirements, we require education and research. Meeting the separate but related challenges of delivering sustainability by the intelligent use of ICT, whilst maximizing ICT's own sustainability is not a simple task.



CASE STUDY IN AFRICA

- African Union (AU) adopted Flagship program in Africa named Agenda 2063
- The prime objective is to *"Address the root causes of conflicts in Africa and strengthen the capacity to prevent and resolve conflicts peacefully in particular using African local values and endogenous practices of a culture of peace"*.
- The expected outcome— mutual understanding and social cohesion; increased and enhanced cultural interactions; highlight the cultural practices of Africa in promoting reconciliation, social cohesion and peace; sustainable use of ecosystems shared by States; establish joint initiatives; use of community radios promoting intercultural/intergenerational dialogue and social cohesion; supporting the African Union's "Make Peace Happen" campaign etc.



Monitoring progress on effective development cooperation principles

The global monitoring framework, consisting of 10 indicators, tracks progress on implementation of the Busan principles and commitments for effective development cooperation. The indicators can be grouped as follows:

Ownership and results

- Use of country results frameworks (indicator 1)
- Aid on budget (indicator 6)
- Quality and use of country systems (indicator 9)
- Untying aid (indicator 10)

Inclusive partnerships

- Enabling environment for civil society organizations (indicator 2)
- Private sector engagement (indicator 3)
- Gender equality (indicator 8)

Transparency and accountability

- Transparency (indicator 4)
- Predictability (indicator 5)
- Mutual accountability (indicator 7)

Global Progress

The 2014 progress report reveals that commitment to the Busan principles by all stakeholders remains strong. Progress on the use of country systems is sustained, but alignment by development providers with national priorities needs to improve. There is a marked improvement in inclusiveness, and greater engagement of nonstate actors. There is progress on transparency and accountability, but they need to be geared toward countries' needs.

Progress of Asia and the Pacific developing member countries

The progress of the region is mixed and varies across the 14 developing member countries that participated in the 2014 survey. Overall, the region has achieved progress in predictability and untying of aid. Accelerated action is needed on the use of country systems, and inclusive partnerships, including on gender equality.

Global Partnership-related indicators in ADB's corporate results framework

- **Use of country results frameworks.** Almost all ADB operations in 2013 explicitly demonstrated links to country sector results, exceeding the 90% performance standard.
- **Use of country systems.** In 2013, 67% of ADB's sovereign operations used developing member countries' public financial management systems, while 33% used country procurement systems.

Source: Organisation for Economic Co-operation and Development (OECD) and United Nations Development Programme (UNDP). 2013. *Guide to the Monitoring Framework of the Global Partnership*. Paris; OECD/UNDP. 2014. *Making Development Co-operation More Effective: 2014 Progress Report*. Paris; and ADB. 2014. *2013 Development Effectiveness Review*. Manila.





SUSTAINABLE HAPPINESS

Lecture: 60



Prof. Atasi Mohanty

Rekhi Centre of Excellence for the Science of Happiness

Sustainable Happiness



***Always remember
to be happy
because you
never know
who's falling in
love with your smile.***

➤ Most people have the same monotonous routine every day, leading to moments of frustration, anger and hopelessness. The ecological issue necessitates the development of innovative solutions to turn our civilization into a more sustainable one swiftly.

➤ **NEED**

1. **Exhaustion**- The monotonicity of today's world has left us exhausted. People are usually overburdened with much work which leaves them with a feeling of sadness and sometimes isolation.
2. **Expectation**- As we know that expectation is the root cause of all suffering. People expect way too much, and when they do not get what they expected, it leaves them unsatisfied.



➤ Sustainable happiness can lead to a more **sustainable lifestyle** and greater **life satisfaction**—and sometimes it's right at our fingertips. Other times it involves reawakening to the joy that comes from simple pleasures or generating options for a lifestyle change.



DR. CATHERINE O'BRIEN

"SUSTAINABLE HAPPINESS IS HAPPINESS THAT CONTRIBUTES TO INDIVIDUAL, COMMUNITY, OR GLOBAL WELL-BEING WITHOUT EXPLOITING OTHER PEOPLE, THE ENVIRONMENT, OR FUTURE GENERATIONS."



➤ Sustainable happiness and well being

Happiness and well-being are integral to building sustainable futures

Focusing only on the goal is difficult, it's not how we are wired to think, and can lead to distress and anxiety

For the long haul, there are some habits of mind which will help to stay on track and achieve sustainable happiness



3 Simple Ways to Sustainable Happiness

- 1 Relax your nervous system
- 2 Be tender with yourself
- 3 Practice happiness for other people's happiness



❑ Sustainable happiness is happiness that contributes to individual, community and/or global well-being without exploiting other people, the environment or future generations. (O'Brien, 2005)

Happiness = ?

Before we chase it,
let's define it clearly.



www.6seconds.org

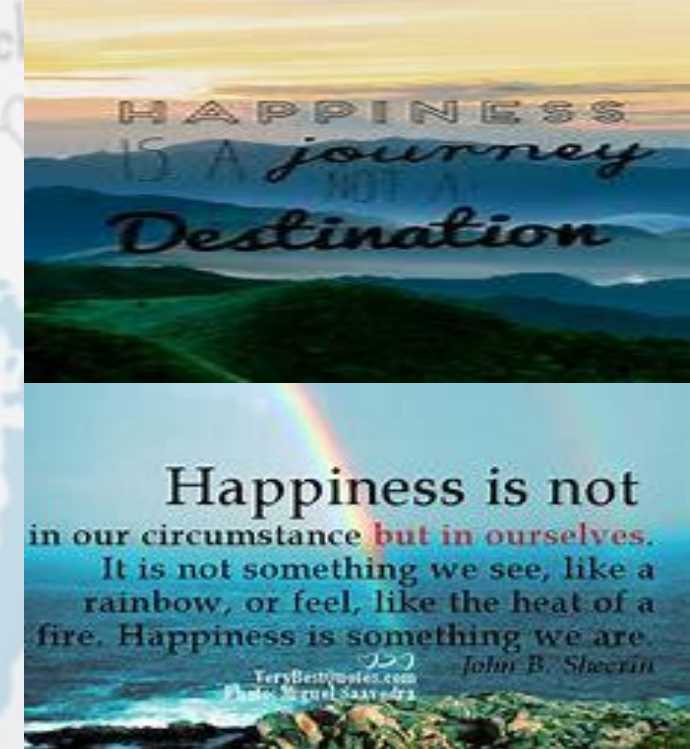


- One's subjective experience of happiness corresponds with numerous positive health outcomes (Seligman et al., 2005);
- Including lower blood pressure, the inclination to seek out and act on health information, and more robust immune systems .
- 1. Happiness protects our heart & mind...
- 2. Happiness strengthens our immune system ...
- 3. Happiness combats stress ...
- 4. Happy people have fewer aches and pains ...
- 5. Happiness combats disease and disability ...



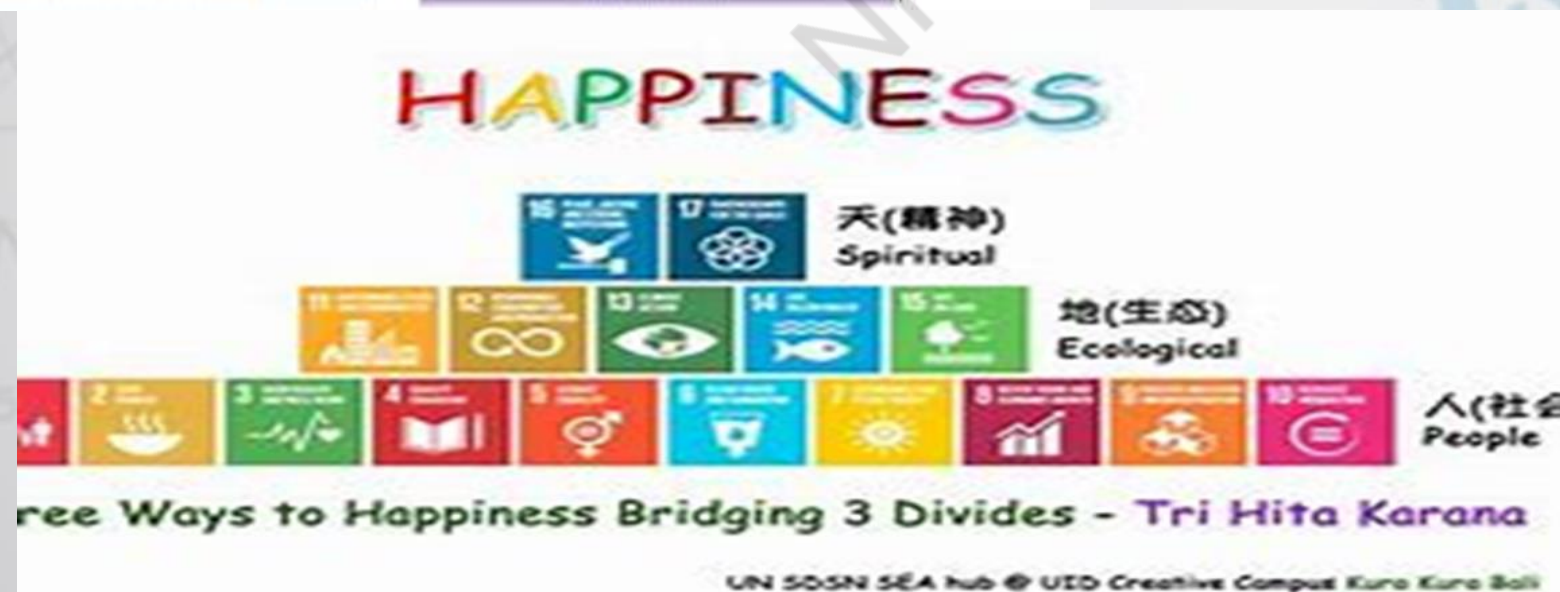
- Happiness and Sustainability
- Happiness is at the heart of who we are and what we do...

In a consumer society, where consumption and happiness are inextricably linked, individuals confuse the “path to the ‘good life’ as the ‘goods life’ .



- ✓ Aim of combining the two terms are: to link happiness to sustainability, now and into the future; to emphasize the reality of our mutual interdependence...
- ✓ Sustainable happiness is a concept that can be used by individuals to guide their actions and decisions on a daily basis...
- ✓ At the community level, it reinforces the need to genuinely consider social, environmental and economic indicators of well-being so that community happiness and well-being are sustainable...
- ✓ At the national and international level it highlights the significance of individual and community actions for the well-being of all – now and into the Future...









SUSTAINABLE HAPPINESS

Lecture: 61



Prof. Atasi Mohanty

Rekhi Centre of Excellence for the Science of Happiness

Subjective Wellbeing



Personality
Pattern

Work-life
Balance

Social
Connections

Challenges and
Diversity of
Tasks

**HAPPINESS
AT WORK**

Autonomy and
Decision making

Acceptance and
Empathy

Financial
Security

Scope for
innovation and
creativity



□ Sustainable happiness= Human Flourishing + Ecological Flourishing

Sustainable Wellbeing challenge activities:

1. Social Connection
2. Mindfulness
3. Lifelong Learning
4. Helping Others
5. Positive Engagements (Physical activities)

Pedagogy & Practice of Sustainable Happiness:

1. Critical Reflection
2. Transformative learning
3. Immersive Learning/Deep Learning/Experiential Learning
4. Deep Contemplation



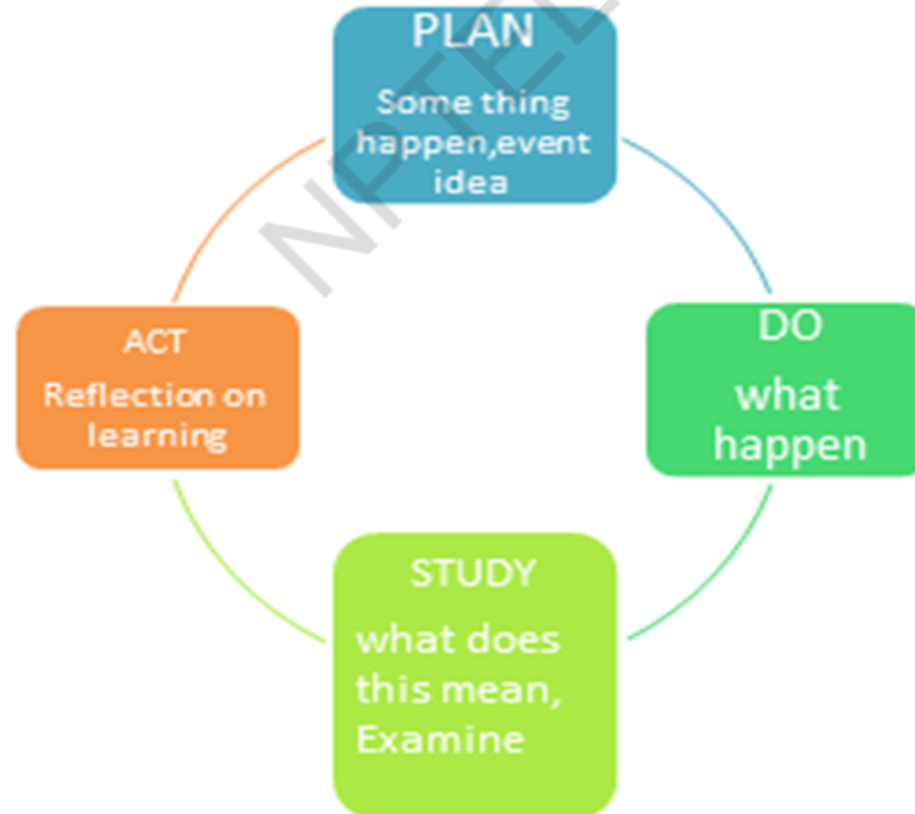
➤ Sustainable Happiness Skills:

1. **Mindfulness:** When one is aware of one's current thoughts, feelings, and surroundings without judging oneself.
2. **Human Connection:** The relationships that exist with significant others, friends, and peers; Strong bonding/relationships are crucial to happiness and well-being,
3. **Gratitude:** When one is thankful and aware of their privilege and identifies the sources of goodness that exist outside of themselves.
4. **Positive Outlook:** When one is positive, one is able to imagine a prosperous outcome and motivate themselves to strive for that positive experience
5. **Purpose:** Having a sense of purpose can assist in contributing to sustainable happiness.
6. **Generosity:** is the desire and ability to assist and be kind to others.



➤ CORE SUSTAINABLE COMPETENCIES

- Anticipatory competency
- System Thinking
- Normative thinking
- Strategic thinking
- Collaborative thinking
- Self aware competency.
- Critical thinking



ROLE OF REFLECTIVE THINKING FOR DEVELOPING SUSTAINABLE COMPETENCIES

- Major intellectual transition between childhood to adulthood.
- It provide skill for mental process learning
- It helps to shape their own thought process
- Helps to modify thought based upon old experience and new learning
- Contd.....

GIBB'S REFLECTIVE THINKING MODEL

Gibbs' Reflective Cycle



- Reflective thinking helps learners develop higher-order thinking

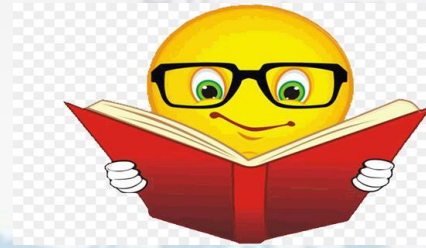
VARIOUS METHODS OF TEACHING REFLECTIVE LEARNING

- This process helps to build sustainable citizens

- Engage constructively and responsibly in sustainable problem solving activity

- Help to improve self awareness

- Uses comparative logical approach



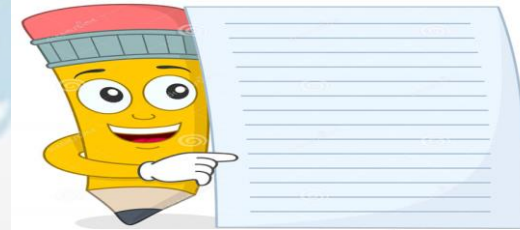
Record

Responding

Relating

Reasoning

Reconstruction



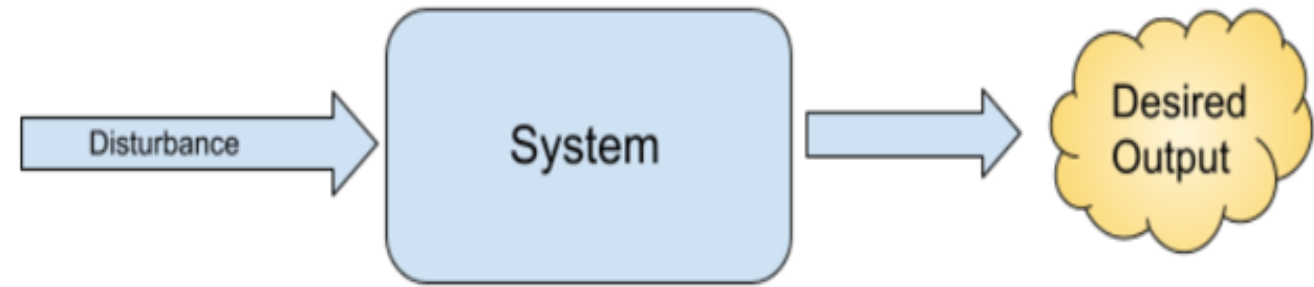
Resilience

- Resilience is the ability of a system to absorb disturbances & retain its basic function and structure
- Disturbance a system can absorb & still remain in the same state.
- Degree of self-organization of the System.
- The ability to build and increase the capacity for learning and adaptation

Resilience is in direct relation with the sustainable competencies mentioned below-

- Systems thinking competency
- Anticipatory competency
- Strategic competency
- Collaboration competency
- Self-awareness competency.

Resilience



Output $\sim k \times$ Disturbance

where; Resilience $\sim 1/k$



Understanding with 3 Facet of Resilience

Resilience concepts	Characteristics	Focus on	Context
<i>Engineering resilience</i>	Return time, efficiency	Recovery, constancy	Vicinity of a stable equilibrium
<i>Ecological resilience</i>	Buffer capacity, withstand shock, maintain function	Persistence, robustness	Multiple equilibria, stability landscapes
<i>Social-ecological resilience</i>	Interplay disturbance and reorganization, sustaining and developing	Adaptive capacity transformability, learning, innovation	Integrated system feedback, cross-scale dynamic interactions

Source: Folke (2006)



Ways to Improve Resilience

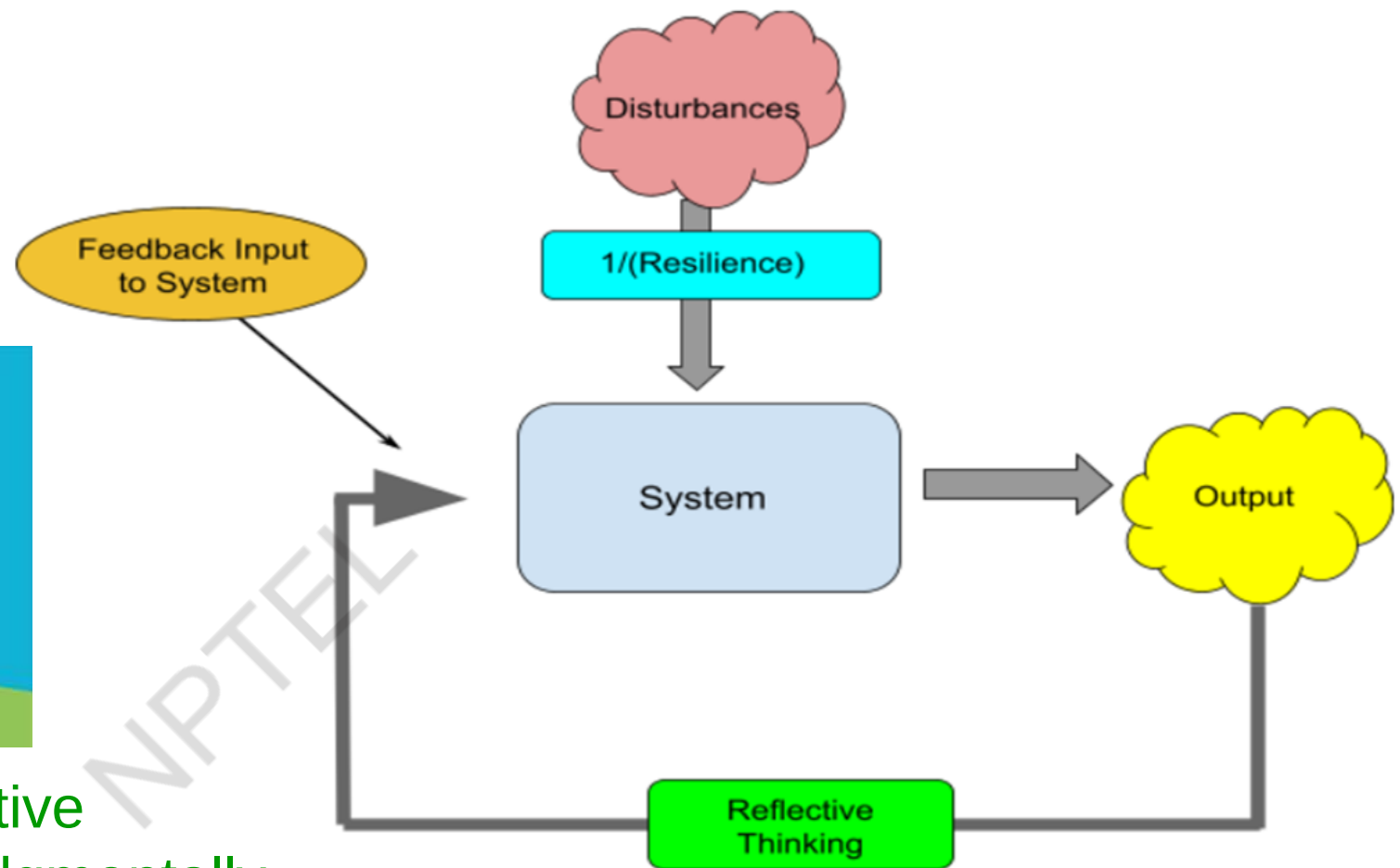
- Develop a social network
- Build optimism.
- Practice self-care
- Find a sense of purpose
- Embrace change.
- Having a broader perspective of life
- Develop goals and problem-solving skills



What is Mindfulness ???



- ✓ Being present and attentive purposefully and non-judgmentally “in the moment”
- ✓ Its focus is to “accept” and “allow”, what is going on in the present, without an immediate pressure to change it.



MINDFULNESS



■ Mindfulness engages practitioners by acclimating them to the present moment's experience realm and developing profound insight into how the mind causes pain.

■ The search of mindfulness via yoga as a practice is a knowledge quest for understanding oneself

■ Hatha yoga, pranayama breathwork, and meditation are all emancipatory discourses that allow long-term enjoyment and well-being



MINDFULNESS STAGES

STAGE 1

Practice begins with pranayama



STAGE 2

Continues with Hatha Yoga emphasizing on controlled breathing and posture



STAGE 3

Sitting, lying or standing in relaxing position



STAGE 4

While listening to some relaxing music begin with meditation for 10-15 minutes



FEEDBACK (LOOP)

- How has mindfulness practice influenced the way you interact with others?
- Why do you use them at that particular time of day?
- What changes have you noticed in your life since you implemented mindfulness practice?
- How do you deal with conflict and anxiety through mindfulness practice?
- Can you elaborate on the general feelings you experience from the practice?



SCHOOL TRAVEL PLANNING

A pan-Canadian project to promote active travel (walking and cycling) on the school journey

The key stakeholders include the school administrator, parent representatives, municipal staff, policing agencies, and public health representatives



Data collection and meetings with all relevant stakeholders who influence or have an interest in school travel

More than 5,000 baseline family surveys provided data on travel modes, barriers to active travel, and sustainable happiness



Even for as little as 15 minutes a day, mindfulness practice has been found to alter one's sense of self and one's sense of self in connection to others, promoting a happier, healthier environment

Napoli (2004) investigated mindfulness with teachers and discovered that these individuals experienced an improvement in all elements of their educational practice.

According to the Garrison Institute (2015), recent developments in neuroscience show that mindfulness practice helps to facilitate awareness and self-regulation while also fostering a calm and focused mind



02

RESULT : MINDFULNESS

Using transportation as an example, it is clear that there are several decision moments during the day when individuals may make decisions that contribute to individual, communal, or global well-being

According to research conducted, One Grade 4 student, who had never heard of sustainable happiness, drew a picture that eloquently captures the concept: “Walking to school makes a happy Earth, which makes happy faces”.

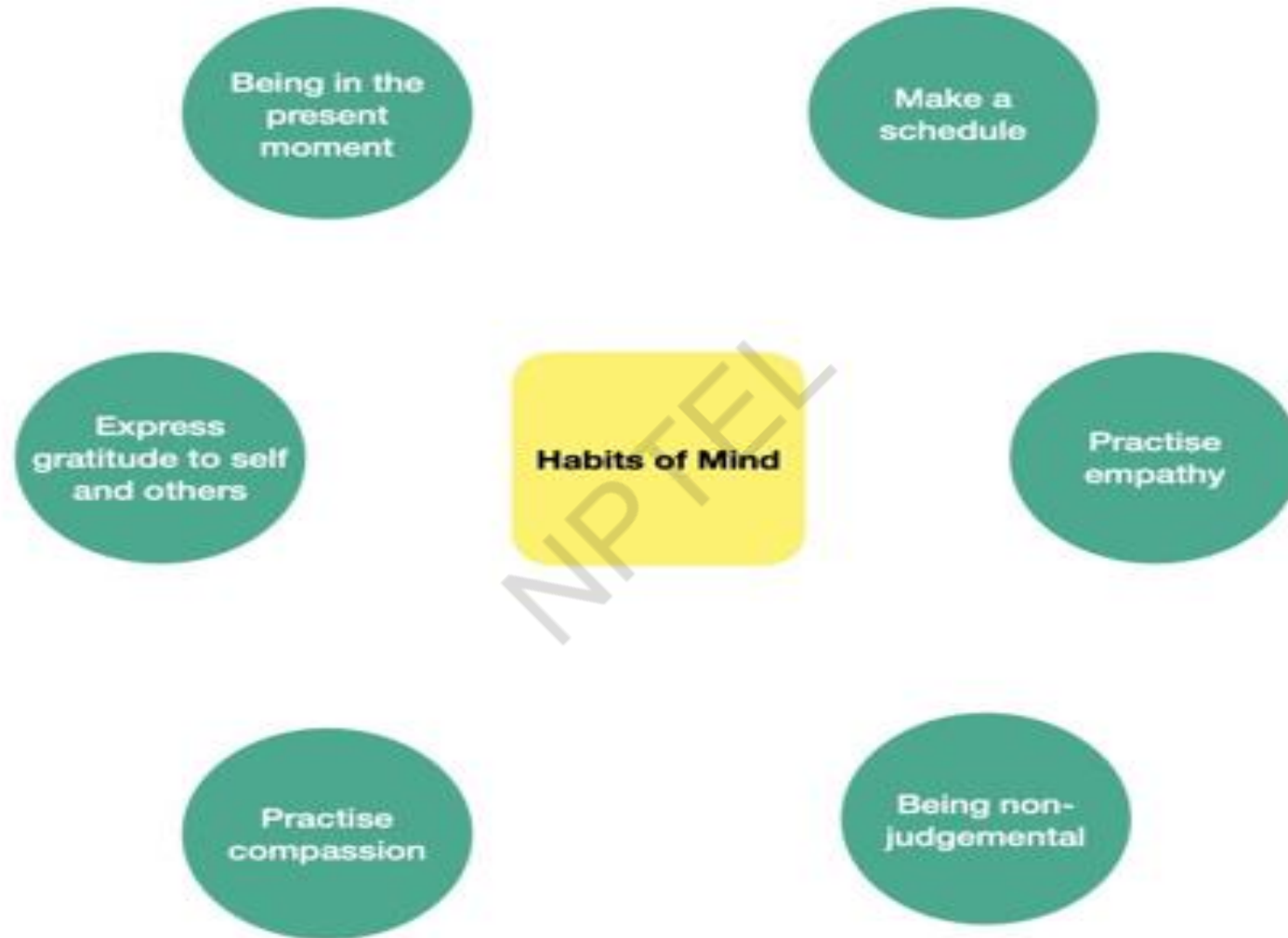


02

RESULT : STP



Habits of Mind to Promote Happiness



Habits of mind that can contribute towards the intentional activities that are most influential for happiness

- Listening with understanding and empathy
- Thinking flexibly
- Thinking Interdependently
- Thinking and communicating with clarity and precision
- Thinking About own thinking
- Helping Nature



Great Benefits of Mindfulness

- #1 – Being mindful of your thoughts and emotions promotes well-being
- #2 – Being mindful can improve your working memory
- #3 – Mindfulness acts as a buffer against the depressive symptoms associated with discrimination
- #4 – Mindfulness can help you make better use of your strengths
- #5 – Mindfulness practice can raise your happiness set-point
- #6 – Mindfulness can make you more resilient
- #7 – It shrinks the stress region in your brain



Conclusion

The union of sustainability, happiness, and well-being can transform individuals, communities and nations, and our planet. It can accelerate shifts in attitudes, policies, practice and behavior.

Case studies of sustainable communities could investigate the association between sustainable lifestyles and life satisfaction



